

Radio-wave propagation modelling over rough sea surfaces and inhomogeneous atmosphere

Modellering av radiovågutsbredning över ojämna vattenytor och inhomogen atmosfär

Månz Nilsson

Faculty of Health, Science and Technology

Master of Science in Engineering Physics

30 hp (ECTS)

Supervisor: Thijs Jan Holleboom

Examiner: Lars Johansson

Date: June 16, 2021

Abstract

The fifth generation of wireless network has been foreseen as a promising candidate to achieve long-range communication links in marine environments. Over watered bodies, evaporation of water causes the atmospheric index to decrease with height. This leads to the formation of atmospheric ducts. Inside the ducting layer, waves propagate in wave-guide like manner and beyond line of sight propagation is possible. To investigate the feasibility of long range radio links, wave propagation modelling is essential. This work presents a mathematical MATLAB model that describes how electromagnetic fields propagates in troposphere. The model is based on the parabolic equation and is discretized via a split-step Fourier transform method. The developed model has been used to study how the propagation is influenced by non-range varying atmosphere, range varying atmosphere, irregular terrain, and lossy boundary conditions. The range varying atmosphere is obtained through linear interpolation of tables with values of the modified refractivity at given heights and ranges. The sea surface consist of large-scale roughness's, modelled as terrain and generated through a random process and small-scale roughness's which are incorporated into the model via the Miller-Brown roughness reduction factor. The model is validated and the results agrees well with literature.

Sammanfattning

Den femte generationens trådlösa nätverk är förutspådd att vara passande för att uppnå långdistanskommunikationskanaler i marina miljöer. Över vatten förångas vatten vilket leder till att atmosfäriska brytningsindexet avtar med höjd. Detta leder till att atmosfäriska kanaler bildas. Inuti kanallagret propagerar vågor som fångas likt vågor i en vågledare och gör det möjligt för vågutbredning bortom synfältet. För att undersöka möjligheten att uppnå lång distans radiolänkar är vågutbrednings modeller ett viktigt verktyg. Det här arbetet presenterar en matematisk MATLAB modell som beskriver hur elektromagnetiska fält propagerar i troposfären. Modellen är baserad på den paraboliska ekvationen och diskrediteras via en Split-step Fourier transformationsmetod. Modellen som har utvecklats has använts för att studera hur vågutbredning påverkas av; icke-avståndsvarierande atmosfär, avstånds varierande atmosfär, oregelbunden terräng och lossa randvillkor. Det avstånd varierande atmosfären erhålls genom linjär interpolation av tabeller innehållandes värden på det modifierade retroaktiviteten vid givna höjder och avstånd. Vattenytan består av storskaliga strävheter och modelleras som terräng och genereras via en slumpmässig, samt småskaliga strävheter som inkorporeras via Miller-Brown roughness reduction factor. Modellen verifieras och resultaten överensstämmer bra med litteratur.

Acknowledgement

First, I want to thank my supervisior Thijs Jan Holleboom for the help and guidance that you provided during the course of this project. Secondly i would like to thank Per Borg, Bengt Hallinger, Tim Lindquist, and Tieto-EVRY for giving me the opportunity to write my master thesis in cooperation with you. Thank you Bengt for the support and help during this project. Thank you Tim for the fruitful discussion and great advices that you have given me during these weeks. Lastly, I would like to thank my family and friends for all the support you have given during the last five years.

Abbreviations

b-LOS - Beyond Line of Sight
DMFT - Discrete Mixed Fourier Transform
EM - Electromagnetic
LOS - Line of Sight
PE - Parabolic Equation
PEC - Perfect Electric Conductor
PLSM - Piecewise Linear Shift Map
SPE - Standard Parabolic Equation
SSFT - Split Step Fourier Transform
SSFT-PE - Split Step Fourier Transform-Parabolic Equation

Contents

1 Introduction **1**

2 Theory **5**

2.1 Wireless Network Systems 5
2.2 Electromagnetic waves 6
2.2.1 Boundary Conditions 8
2.3 The Parabolic Equation 10
2.3.1 Basic Derivation 11
2.3.2 Split Step Fourier Transform 14
2.3.3 Range Dependence 19
2.3.4 Wide angle Split Step Solution 20
2.4 Wave equations for propagation over the Earth 22
2.4.1 Parabolic equation with Spherical coordinates 22
2.4.2 Transformation to flat earth 26
2.4.3 Path Loss 29
2.5 Boundary conditions 31
2.5.1 Ground 31
2.5.2 Discrete Mixed Fourier Transform 33
2.5.3 Initial condition 35
2.5.4 Domain Truncation 36
2.6 The Atmosphere 36
2.6.1 Refractive Index 37
2.6.2 Ducts 40
2.7 Terrain Modelling 42
2.7.1 Ground 42
2.7.2 Ocean 46

3 Method **53**

4 Results and Discussion **57**

4.1 Propagation in Free space 57
4.2 Ducting 63
4.2.1 Range Independent 64
4.2.2 Range Varying surface duct 67
4.3 Terrain 68
4.4 Sea surface 70

4.5 The Model 75

5 Conclusion **79**

6 Future Work **81**

Appendices **87**

A **87**

List of Figures

1 Illustration of a simple network system 5
2 Illustration of reflection and transmission of waves at an interface 9
3 The domain of interest and propagation direction 12
4 Interpretation of the splitting operator 18
5 Spherical coordinates 23
6 The flat earth transform 27
7 Ray trajectory's in the atmosphere 39
8 M-profiles as function of altitude 41
9 Illustration of the staircase approach of terrain modelling . . . 43
10 Marching solution over staircase terrain 44
11 Elfouhaily sea wave spectrum 51
12 Generated sea surface profiles 51
13 Flowchart for the algorithm used in this project. 55
14 Propagation in free space with a horizontally polarized field . 58
15 Propagation in free space with a vertically polarized field . . . 59
16 Horizontally polarized field propagating in standard atmosphere 59
17 Horizontally polarized field propagating in standard atmosphere with modified environmental propagator 60
18 Horizontally polarized field propagating in standard atmosphere 61
19 Horizontally polarized field propagating in standard atmosphere 62
20 Vertically polarized field propagating in standard atmosphere . 62
21 Vertically polarized field propagating in standard atmosphere . 63
22 Path loss coverage diagrams for propagation in a bilinear surface duct 65
23 Path loss as function of height and path loss as function of range for propagation in a surface ducts 66
24 Ducting path loss comparison with PETOOL v.2 66
25 Propagation in range varying atmosphere 68
26 Propagation over a wedge terrain 70
27 Propagation over rough sea surface 72
28 Propagation over generated sea surface 73
29 Path loss coverage diagrams for propagation over sea surfaces in a bilinear surface duct 75

List of Tables

1 A comparison between the narrow- and wide-angle propagators. 21
2 Atmospheric conditions and the corresponding modified refractive index gradient where the units are in M-units/km [1]. 40

1 Introduction

During the last couples of decades mobile wireless communication systems has steadily evolved from first to fourth generation wireless network. The number of mobile subscribers has increased exponentially since the release of the first cellular wireless network in 1979. The development of smart devices and multimedia applications has led to a rapid increase in data traffic. New, faster and low latency wireless networks must be developed to support the expected increase of data traffic in the coming decade. Today, society stands in the middle of a transition period in going from the fourth generation to fifth generation (5G) cellular wireless network.

Over-seas industries such as export of trade goods, oil exploitation, fishery, tourism and environmental monitoring have expanded during the last decades and so has maritime military. With increased naval activity, the data traffic have quickly grown and current network systems are not able to provide the data rates needed or stable connections. Especially at long distances of the shore. Existing maritime networks are using High Frequency (HF), Very High Frequency (VHF) and Ultra High Frequency (UHF) broad band systems to communicate over short distances and satellite systems for long-range communications. Satellite systems suffers from high implementation and maintenance cost, large delays, and are prone to hostile jamming [2]. Furthermore, the broad band systems are not able to provide the high data rates that are demanded. Thus, developing land-based network systems are of great interest, especially from a military point of view. The terrestrial mobile networks that today are being used have the wanted capacity and thus it is desired for maritime users to access inland network systems. During initial exploitation of new network systems, field propagation path loss models derived from either empirical measurements, semi-empirical- or analytical methods are often utilized for wireless network planning [3]. The path loss models are used to estimate signal attenuation as a function of range, terrain profile and carrier frequency. For marine environments both empirical and analytical path-loss models have been developed where most short-range models are empirically based [4][5]. The long-range models are often based on analytical methods such as the one developed in [6][7]. Furthermore, organisations such as the third generation partnership program (3GPP) and the international telecommunication union (ITU) have presented path-loss model that supports distances up to 100 km, and 1000 km respectively [8][9].

Despite this the 3GPP and researchers that tested the ITU model have been underscoring the need to complete these models by further research [10].

The use of land-based network systems to establish long-range radio links comes with some problems. To establish connections it is required that the trajectory of the radio waves follow the curvature of the earth to reach a specific receiver located close to the surface of earth. In standard atmosphere, propagating waves are refracted downward towards earth's surface with a curvature greater than earth's radius [11]. Field strengths are therefore reduced past the horizon resulting in large signal attenuations. Thus, in standard atmospheric conditions the range of communication is restricted to radio line of sight (LOS) which is approximately 40 km. However, in special conditions, the atmospheric refractive index rapidly decreases with height causing propagating waves to be refracted downwards towards earth's surface with a curvature less than earth's radius [11]. This phenomenon is known as ducting and can cause waves to propagate beyond the line of sight (b-LOS). Maintaining stable long-term, long-range radio links therefore require constant ducting activity. Atmospheric ducts are formed by rapid variations of the atmospheric refractive index which occur when atmospheric conditions such as temperature, pressure, wind and most importantly humidity changes. Over watered bodies where there is constant evaporation of water, the humidity of the air just above the surface increases and gives rise to evaporation ducts which are almost permanent in marine environments [2]. In presence of evaporation ducts b-LOS radio links have been shown to possible [12][13][6][7]. In [12] and [13] the authors employed analytical models based on the well established theory of parabolic equation (PE) for radio wave propagation simulations and found that minimal power drop occurred for carrier frequency of 10.5 GHz. Based on these results a large scale signal attenuation model for carrier frequencies ranging from 5-15 GHz was developed in [6]. Since the 5G network utilizes a previously unused frequency spectrum ranging from 3$\sim$300 GHz [14] it has been foreseen as a good candidate to achieve stable b-LOS radio-links over-seas [4]. Even though the developed analytical models demonstrate the feasibility of long-range radio-links they tend to under predict power drops when compared to measurements [12] and need to be further improved. For example, neither [12], [13] or [6] consider signal attenuation due to horizontally inhomogeneous atmosphere that can arise in coastal areas where air-sea masses interchange causing variation of ducting characteristics [15]. Furthermore sea waves are not modelled at all

which presence affects the signal attenuation [16]. Sea surfaces has roughness features that are often included in an effective reflection coefficient, which are often calculated by using the Miller-Brown approximation [17]. This model is however dependent on local grazing angle which is hard to predict using PE methods. Furthermore, it does not account for shadowing. Benhmammouch et al. presented a method that models large scale roughness's as a staircase surface which height profile depends on the Elfouhaily sea surface spectrum [18][19]. The small scale roughness's are modelled as sea surface roughness's through the Miller-Brown approximation [19]. The model developed by Benhmammouch et al. in [19] fails to accurately model capillary waves. Therefore models like the improved fractal model have been developed [20]. However, most studies using fractal sea surface uses finite difference (FD) based PE method which increases computational times for large domains.

The PE method was first introduced by Leontovhich and Fock in 1946 as a way of solving atmospheric propagation problems. Researchers have over the years developed multiple advanced wave propagation prediction tools based on PE methods. The advanced refractive effects prediction system (AREPS) based on the work of Barrios [21] was originally developed for the U.S navy. the tropospheric electromagnetic parabolic equation routine (TEMPER) [22] has been developed by the John Hopkins University Applied Physics Laboratory. Lastly, Ozgun et al. developed the parabolic equation tool (PETOOL) and PETOOL V.2 [23][24]. However, neither AREPS or TEMPER are publicly available. Even though PETOOL and PETOOL V.2 are free MATLAB based toolboxes, their description of terrain is rather rudimentary and tends to overestimate fields strengths behind objects. Lindquist developed a model that approximated terrain features through the piecewise linear shift map method [25]. This is considered as a better approximation of terrain features than the one used in PETOOL.

This thesis serves as a final report for a degree project for a Master of Science in Engineering Physics. The project was performed at TietoEVRY in Karlstad between January and June 2021.

Aim

The work in this thesis aims to develop an numerical model in MATLAB, based on analytical methods that can be used to investigate long-range radio-wave propagation in marine environments. The model will be a continuation of the work by Lindquist in [25] and further development shall include;

- Both height and range dependent atmospheric refractive index effects.
- A sea wave modelling and effects such as electromagnetic wave scattering and reflection of a rough sea surface.
- Estimation of signal attenuations by implementing PE algorithms.

The domain of calculation will be up distances up to 500 km and with carrier frequencies ranging from 0.1-15 GHz. Numerous methods of implementing the above-mentioned effects have been developed over the years and in order to determine which is best suited for this specific application a literature study will be performed.

Layout

The thesis is organized as follows. In section 2, The theory of; parabolic equation and wave propagation, atmospheric effects, sea wave modelling, and ray tracing is presented. In section 3 the working principle of the developed model is explained. In section 4 results from sea wave modelling, wave propagation- and ray tracing simulations are presented together with a discussion of the result. In section 5 a conclusion is presented, where the outcome of the work is summarized. Section 6 closes the thesis with and outlook and possible future studies.

2 Theory

2.1 Wireless Network Systems

Wireless network systems transmit and receive data through wireless channels. They do so by employing transmitting antennas that accelerate charges causing radiation of electromagnetic waves into space. The radiated waves propagate towards a receiving antenna that processes the signal. In figure 1 a simple configuration of a cellular network system is illustrated where waves from a transmitter propagate either directly or through diffraction to the user equipment (UE) device with a receiving antenna.

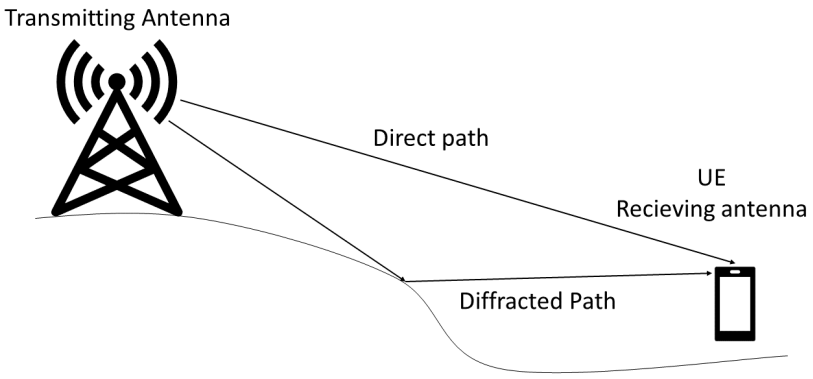


Figure 1: Schematic illustration of a simple network system. Rays propagates from an antenna and at the receiver the signal is often made up of both diffracted and direct ray paths.

Radio-waves that propagates in space interact with surrounding such as buildings, atmosphere, and underlying terrain. The interaction causes refraction, reflection, and diffraction of waves that reduce the energy reaching the receiver. The ratio between the transmitted signal power and the received signal power is known as path-loss (PL), and is in general expressed as:

$$PL = 10 \log_{10} \left(\frac{P_t}{P_r} \right) \tag{1}$$

where P_t and P_r are the transmitted and the received powers, respectively, and PL is in decibel dB. During the initial exploitation of wireless network systems, path-loss models are often used to estimate signal attenuations, carrier frequencies, and antenna heights. Thus, reliable path-loss models are essential for the efficient planning of wireless networks.

2.2 Electromagnetic waves

Electromagnetic fields (EMFs) consist of space and time-dependent electric and magnetic vector fields denoted as $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{H}(\mathbf{r}, t)$, respectively. In 1861 James Clerk Maxwell published a paper in which he presented a set of equations that later were reduced to four partial differential equations that describe how EMFs are generated by changes in the fields, currents, and charges. These are today known as Maxwell's equations. Their differential form reads as follows

$$\nabla \cdot \mathbf{D}(\mathbf{r}, t) = \rho(\mathbf{r}, t) \tag{2}$$

$$\nabla \cdot \mathbf{H}(\mathbf{r}, t) = 0 \tag{3}$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t} \tag{4}$$

$$\nabla \times \mathbf{H}(\mathbf{r}, t) = \mathbf{J}(\mathbf{r}, t) + \frac{\partial \mathbf{D}(\mathbf{r}, t)}{\partial t} \tag{5}$$

where $\mathbf{D}(\mathbf{r}, t)$ is the displacement field, $\mathbf{B}(\mathbf{r}, t)$ is the magnetic flux, $\rho(\mathbf{r}, t)$ is the charge density and $\mathbf{J}(\mathbf{r}, t)$ is the current density. Together with the Lorentz force, these equations are the foundation of electrodynamics. By applying suitable boundary conditions these equations can be solved uniquely [26].

By assuming that the domain of interest is current, and source free (i.e $\rho(\mathbf{r}, t) = 0$ and $\mathbf{J}(\mathbf{r}, t) = 0$) Equation (2) - (5) reduce to

$$\nabla \cdot \mathbf{D}(\mathbf{r}, t) = 0 \tag{6}$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0 \tag{7}$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t} \tag{8}$$

$$\nabla \times \mathbf{H}(\mathbf{r}, t) = \frac{\partial \mathbf{D}(\mathbf{r}, t)}{\partial t} \tag{9}$$

Further assuming that the medium inside the domain is linear and homogenous the relation between displacement vector and electric field is $\mathbf{D}(\mathbf{r},t) = \epsilon\mathbf{E}(\mathbf{r},t)$, and the relation between magnetic flux density and magnetic field is $\mathbf{B}(\mathbf{r},t) = \mu\mathbf{H}(\mathbf{r},t)$, where ϵ and μ is the permittivity and permeability of the medium respectively. Equations (6)-(9) become

$$\nabla \cdot \mathbf{E}(\mathbf{r},t) = 0 \tag{10}$$

$$\nabla \cdot \mathbf{H}(\mathbf{r},t) = 0 \tag{11}$$

$$\nabla \times \mathbf{E}(\mathbf{r},t) = -\mu\frac{\partial \mathbf{H}(\mathbf{r},t)}{\partial t} \tag{12}$$

$$\nabla \times \mathbf{H}(\mathbf{r},t) = \mu\epsilon\frac{\partial \mathbf{E}(\mathbf{r},t)}{\partial t}. \tag{13}$$

These four equations are coupled by taking the curl of equation (12) and (13) and using the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}$ they can be decoupled

$$\nabla^2\mathbf{E}(\mathbf{r},t) = \mu\epsilon\frac{\partial^2 \mathbf{E}(\mathbf{r},t)}{\partial t} \tag{14}$$

$$\nabla^2\mathbf{H}(\mathbf{r},t) = \mu\epsilon\frac{\partial^2 \mathbf{H}(\mathbf{r},t)}{\partial t} \tag{15}$$

In this work only harmonic time dependence will be considered, i.e $e^{-i\omega t}$ where ω is the angular frequency of the fields. Equation (14) and (15) can thus be simplified by taking the differential with respect to time. This yields the following expressions

$$\nabla^2\mathbf{E}(\mathbf{r}) = -\mu\epsilon\omega^2\mathbf{E}(\mathbf{r}) \tag{16}$$

$$\nabla^2\mathbf{H}(\mathbf{r}) = -\mu\epsilon\omega^2\mathbf{H}(\mathbf{r}). \tag{17}$$

Which is reduced to The Helmholtz wave equations by setting $|\mathbf{k}| = k = \sqrt{\mu\epsilon\omega^2}$

$$\nabla^2\mathbf{E}(\mathbf{r}) + k^2\mathbf{E}(\mathbf{r}) = 0 \tag{18}$$

$$\nabla^2\mathbf{H}(\mathbf{r}) + k^2\mathbf{H}(\mathbf{r}) = 0. \tag{19}$$

It is more convenient to express (19) with the refractive index for homogenous medium i.e $n = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} = C$, where μ_0 and ϵ_0 is the permeability and

permittivity of free space respectively. Assuming the atmosphere to be non-magnetic i.e $\mu = \mu_0$ the refractive index is written $n = \sqrt{\epsilon/\epsilon_0}$ and equation (18) and (19) can be expressed as

$$\nabla^2\mathbf{E}(\mathbf{r}) + k^2n^2\mathbf{E}(\mathbf{r}) = 0 \tag{20}$$

$$\nabla^2\mathbf{H}(\mathbf{r}) + k^2n^2\mathbf{H}(\mathbf{r}) = 0 \tag{21}$$

where $k = \sqrt{\mu_0\epsilon_0\omega^2}$ is the wave number in vacuum.

Before closing this part, the assumption that has been made needs to be addressed. The domain in this thesis is the troposphere which refers to the lower part of the atmosphere, which justifies our first assumption since no accumulation of charge or current is present in our atmosphere. Posing full atmospheric homogeneity is not required for deriving the Helmholtz equation. However, if ϵ would have a spatial dependence, terms containing the gradient of ϵ would appear. Variations of permittivity are very small. Above the earth's surface, it rarely exceeds 1.0004 [16]. This thesis will use carrier frequencies ranging from $0.1 - 15$ GHz, corresponding to wavelengths of centimeters. Thus $\nabla\epsilon$ is extremely small compared to the radio-waves wavelength and can therefore be neglected [27]. This means that the derived Helmholtz equation is reasonable to work with if the refractive index has small and smooth variations. Lastly, only considering fields exhibiting harmonic time dependence restricts the work in this thesis to well-behaved fields that are more convenient for calculations.

2.2.1 Boundary Conditions

As fields propagate through space they can travel in different mediums separated by a boundary interface where Maxwell's equations must satisfy a set of boundary conditions at both sides of the interface. Figure 2 illustrates the situation where incoming waves travel from one medium into another. The boundary conditions read as follows [26]

$$\begin{cases} \epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp} \\ E_1^{\parallel} = E_2^{\parallel} \\ \mu_1 H_1^{\perp} = \mu_2 H_2^{\perp} \\ H_1^{\parallel} = H_2^{\parallel} \end{cases} \tag{22}$$

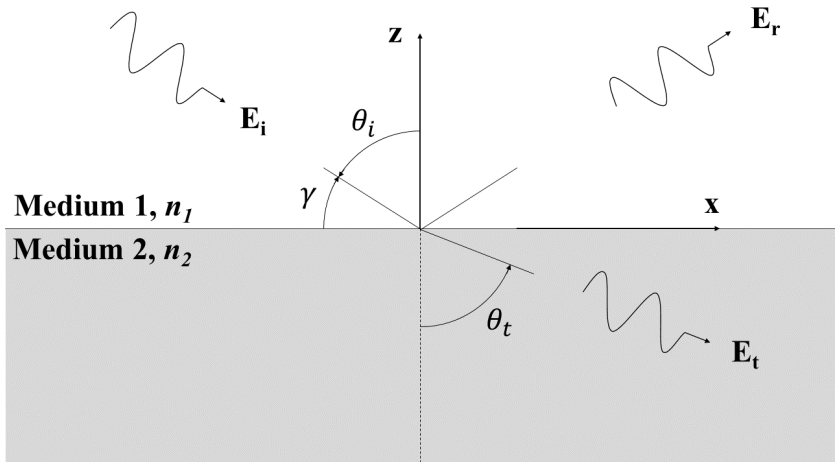


Figure 2: Incident field is reflected and transmitted at the interface between two different medium

where $E^{\perp}$, $H^{\perp}$ is the electric and magnetic field components perpendicular to the interface respectively, and $E^{\parallel}$, $H^{\parallel}$ is the electric and magnetic field components parallel to the interface respectively.

Numerous properties can be deduced from these boundary conditions, assuming that both mediums are non magnetic ($\mu_1 = \mu_2 = \mu_0$) Snell's law reads as follows

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \frac{\eta_2}{\eta_1} = \frac{\sqrt{\epsilon_1}}{\sqrt{\epsilon_2}} \tag{23}$$

where n_1 and n_2 is the refractive index of medium 1 and medium 2, respectively and η_2 and η_1 is the wave impedance for a wave propagating in medium 2 and medium 1, respectively. Snell's law is also known as the law of reflection and tells us that angle of refraction is related to the refractive indices of the two materials [26]. The importance of Snell's law will become apparent when discussing atmospheric propagation in section 2.6.

From the boundary conditions Fresnel's equation can also be derived. Fresnel's equations relates the reflective and the incident fields through Γ

which is defined as [26]

$$\Gamma_{\parallel} = \frac{E_{0r}}{E_{0i}} = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i} \tag{24}$$

$$\Gamma_{\perp} = \frac{E_{0r}}{E_{0i}} = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} \tag{25}$$

where E_{0r} and E_{0i} corresponds to the magnitude of reflective and incident field respectively. The parameters $\Gamma_{\parallel}$ and $\Gamma_{\perp}$ corresponds to the two cases when the incident field is parallel and perpendicular to the plane of interface respectively. However, the form of the Fresnel equation is not suitable for the applications that concern this thesis. It is better to express equation (24) and (25) in terms of the grazing angle γ as seen in figure 2 (i.e the angle between electromagnetic waves and the boundary interface) and the medium impedance η as follows [28]

$$\Gamma_{\parallel} = \frac{\frac{\eta_1^2}{\eta_2^2}\sin\gamma - \sqrt{\frac{\eta_1^2}{\eta_2^2} - \cos^2\gamma}}{\frac{\eta_1^2}{\eta_2^2}\sin\gamma + \sqrt{\frac{\eta_1^2}{\eta_2^2} - \cos^2\gamma}} \tag{26}$$

$$\Gamma_{\perp} = \frac{\sin\gamma - \sqrt{\frac{\eta_1^2}{\eta_2^2} - \cos^2\gamma}}{\sin\gamma + \sqrt{\frac{\eta_1^2}{\eta_2^2} - \cos^2\gamma}}. \tag{27}$$

In section 2.5 equation (26) and (27) will be used for implementing boundary condition for ground material into the model.

2.3 The Parabolic Equation

The Helmholtz equations derived in the previous section has reduced the complexity of Maxwell's equation, despite this, solving the Helmholtz equation is still hard and approximation methods need to be implemented to further simplify the problem. The parabolic approximation has become one of the most popular methods for solving Helmholtz equation. In this section, the theoretical framework of the Parabolic equation will be introduced to the reader by following the work by Levy in [16].

2.3.1 Basic Derivation

The derivation starts by considering a Cartesian coordinate system (x, y, z). Even though this might seem odd since this thesis aims to investigate long-range wave propagation over a spherical surface (earth's surface) which is better represented by spherical coordinates the chosen approach will become apparent later on. The focus will be on solving two-dimensional wave propagation problems where the fields are independent of the transverse direction (y-direction, chosen by convention). Only considering two-dimension allows electric and magnetic fields to be decomposed into horizontally and vertically polarized fields that propagate independently of one another since there are no depolarization effects [16]. The only non-zero EMF components are $\mathbf{E}_y$ and $\mathbf{H}_y$ for horizontal and vertical polarization respectively, and the appropriate field components ψ is defined as

$$\psi(x,z) = \begin{cases} E_y(x,z) & \text{Horizontal Polarization} \\ H_y(x,z) & \text{Vertical Polarization.} \end{cases} \tag{28}$$

As mentioned in the previous section a unique solution requires knowledge and application of boundary conditions. The domain starts at a certain point (x_0, z_0) which refers to the position of the transmitting antenna, and the initial condition is given by the antenna pattern discussed in Section 2.5.3. The domain bottom is the ground and the top boundary generally extends infinitely but when solving the actual problem it needs to be cut off at a certain height with suitable boundary conditions. The domain boundary conditions will be further discussed in 2.5.

By convention, the x-direction is chosen as the paraxial direction. A schematic setup of a typical tropospheric propagation problem is illustrated in figure 3, where a transmitting antenna radiates energy about the paraxial direction inside the cone defined by the dashed lines. Using the convention, the y-direction can be seen to point into the paper. In order for us to develop the wave equation we start by making the assumption that the refractive index $n(x, z)$ varies smoothly inside the domain [16]. From equation (20) and (21) the field component ψ satisfies

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial z^2} + k^2 n^2 = 0 \tag{29}$$

which is known as the two-dimensional scalar wave equation. A reduced wave

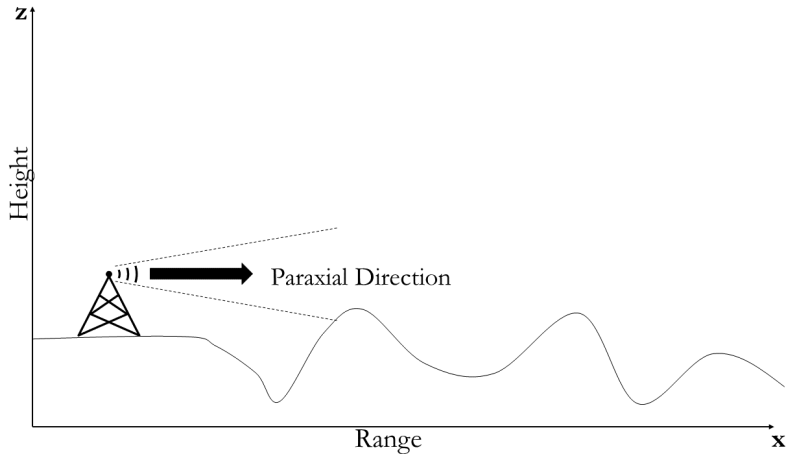


Figure 3: The domain of interest in radio wave propagation problems where the bottom boundary is limited by the ground indicated by the curvy line in the figure. An antenna emit radiation in the paraxial direction. The x-direction follows the surface of the earth and the z-direction is the height above earths surface.

function $u(x, z)$ is introduced an defined as [16]

$$u(x, z) = e^{-ikx}\psi(x, z). \tag{30}$$

The reason for introducing the reduced function is that the energy of the fields is mainly propagating in the paraxial direction and this substitution gives a field u that is slowly varying with respect to x and thus convenient numerical properties are obtained [16]. Inserting equation (30) in equation (31) and differentiating gives

$$\frac{\partial^2 u}{\partial^2 x} + 2ik\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial^2 z} + k^2(n^2 - 1) = 0 \tag{31}$$

Which can be factorized into [16]

$$\left\{\frac{\partial u}{\partial x} + ik(1 - Q)\right\}\left\{\frac{\partial u}{\partial x} + ik(1 + Q)\right\}u = 0 \tag{32}$$

Where Q is known as the Pseudo vector and is defined as

$$Q = \sqrt{1 + q} = \sqrt{\frac{1}{k^2}\frac{\partial^2}{\partial z^2} + n^2(x, z)} \tag{33}$$

and

$$q = \frac{1}{k^2}\frac{\partial^2}{\partial z^2} + (n(x,z) - 1) \tag{34}$$

The factorization in equation (32) might raise some questions. since Q depends on the refractive index and $n(x,z)$ is a function of range, Q does not commute with the range derivative and the factorization is incorrect. However, it is common to assume that the commutation error remain small. This is in fact the case for the scenarios dealt with in this thesis and we will give proof to that later on. For the time being we assume that the errors remains small such that the factorization is considered as valid.

From equation (32) it is clear that two solutions exist, splitting these equations yields

$$\frac{\partial u}{\partial x} = -ik(1-Q)u \tag{35}$$

$$\frac{\partial u}{\partial x} = -ik(1+Q)u \tag{36}$$

where equation (35) and (36) corresponds to forward and backward propagation waves respectively. Exact solutions are given by solving both equations simultaneously with suitable initial and boundary conditions. However, since the main concern in this thesis is long-range propagation the forward solution is enough [29].

Solution to the differential equation (35) is easily obtained and of the form

$$u(x,z) = Ae^{ikx(Q-1)} \tag{37}$$

where A is a constant corresponding to the amplitude of propagating waves. If the initial reduced field $u(x_0, z_0)$ is known the outgoing parabolic equation wave equation have the following solution

$$u(x + \Delta x, z) = e^{ik\Delta x(Q-1)}u(x,z). \tag{38}$$

Thus the forward propagation field at a given range step is obtained by knowing the field strength at a previous range step and knowledge about domain top and bottom boundary conditions which gives the parabolic equation method great numerical properties. However, for actually solving the forward propagation wave function Q must be approximated such that it can be solved with simple algorithms. Thus, equation (38) depends on the form

of Q. A standard approximation of Q is usually made by expanding equation (33) in a Taylor series in q and only keeping first order terms

$$\begin{aligned} Q = \sqrt{1+q} \approx 1 + \frac{q}{2} &= 1 + \frac{1}{2}\left(\frac{1}{k^2}\frac{\partial^2}{\partial z^2} + (n(x,z)-1)\right) \\ &= \frac{1}{2}\left(\frac{1}{k^2}\frac{\partial^2}{\partial z^2} + (n(x,z)+1)\right) \end{aligned} \tag{39}$$

Substituting equation (39) into equation (35) and rearranging the equation gives

$$\frac{\partial^2 u(x,z)}{\partial z^2} + 2ik\frac{\partial u(x,z)}{\partial x} + k^2(n^2-1)u(x,z) = 0. \tag{40}$$

The equation above is known as the Standard Parabolic Equation (SPE) [16]. Since the SPE was derived through an approximation of the pseudo-operator small errors may occur. To evaluate the magnitude of the errors we evaluate the first neglected term of the Taylor expansion of Q. It can be shown to be proportional to $\sin^2\alpha$ where α is the angle of deviation from the paraxial direction [16]. Since the magnitude of the error is given by taking the absolute value squared of the first neglected term the error can be shown to be proportional to $\sin^4\alpha$. For sufficiently small angles ($10^\circ - 15^\circ$) the magnitude of the error is small enough for the SPE to be used [16]. Solving the SPE requires algorithms that will be described in the next section.

2.3.2 Split Step Fourier Transform

When the parabolic equation first was introduced it did not gain the attention that it might have deserved. The problem was still too hard to solve since no known algorithms for solving it was known and computational power was insufficient. This changed when in 1973, Hardin and Tappert presented the Split-Step Fourier Transform algorithm for solving the parabolic equation [30]. The method is a marching algorithm i.e the field strength at a certain range is calculated through the field at the previous step. Since Hardin and Tappert introduced the SSFT-PE, multiple algorithms have been developed to solve the parabolic equation. For long-range tropospheric wave propagation SSFT-PE has been widely used due to its numerical stability when applying large range steps and its ability to be applied for ducting conditions [29].

The SSFT-PE method will now be described by starting at equation (40) and working us through from constant refractive index to range and altitude dependent refractive index. The definition of the Fourier transform is given as

$$\mathcal{F}\{f(x)\} = F(p) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ipx}dx \tag{41}$$

and the inverse Fourier transform as

$$\mathcal{F}^{-1}\{F(p)\} = f(x) = \int_{-\infty}^{\infty} F(p)e^{2\pi ipx}dp. \tag{42}$$

Taking the Fourier transform of $u(x, z)$ in z-direction for each terms in equation (40) yields

$$\mathcal{F}\{u(x, z)\} = U(x, p) = \int_{-\infty}^{\infty} u(x, z)e^{-2\pi ipx}dz. \tag{43}$$

If the refractive index would not be constant the Fourier transform will contain a convolution integral and no simple solution is possible. The Fourier transform of term with second order derivate with respect to z can be evaluated by applying partial integration and the assumption that the reduced field vanishes at $\pm\infty$ thus

$$\mathcal{F}\left\{\frac{\partial^2 u(x, z)}{\partial z^2}\right\} = \int_{-\infty}^{\infty} \frac{\partial^2 u(x, z)}{\partial z^2} e^{-2\pi ipz}dz = -4\pi^2 p^2 U(x, p), \tag{44}$$

and the second term

$$\mathcal{F}\left\{\frac{\partial u(x, z)}{\partial x}\right\} = \frac{\partial U(x, p)}{\partial x} \tag{45}$$

The Fourier transform of equation (40) is thus given by

$$-4\pi^2 p^2 U(x, p) + 2ik\frac{\partial U(x, p)}{\partial x} + k^2(n^2 - 1)U(x, p) = 0. \tag{46}$$

Rearranging the equation and setting first orders derivates on left hand side and zeroth ordered terms on the right hand side yields

$$\frac{\partial U(x, p)}{\partial x} = -\frac{i}{2k}\left(4\pi^2 p^2 - k^2(n^2 - 1)\right) U(x, p). \tag{47}$$

Equation (47) is a first order differential equation which solution is given as

$$U(x,p) = e^{\frac{ik}{2}(n(x,z)^2-1)x} e^{\frac{-4\pi^2 ip^2 x}{2k}} \tag{48}$$

The solution must be given in real space and therefore the reverse Fourier transform is applied

$$u(x,z) = \mathcal{F}^{-1}\{U(x,p)\} \tag{49}$$

To obtain the solution of the SPE from the SSFT, a marching step technique is employed. The field at $u(x+\Delta x)$ is given as

$$u(x+\Delta x) = \mathcal{F}^{-1}\left\{U(x+\Delta x, z)\right\}. \tag{50}$$

Terms that do not contain z can be taken out of the inverse transform and the final expression for the reduced field at the point $x+\Delta x$ is then given as [16]

$$u(x+\Delta x, z) = e^{\frac{ik}{2}(n^2-1)\Delta x} \mathcal{F}^{-1}\left\{U(x,p) e^{\frac{-4\pi^2 ip^2 \Delta x}{2k}}\right\}. \tag{51}$$

The variable p that was introduced in equation (42) can be shown to correspond to a plane wave with wavenumber p. The wave is propagating in the direction described by the angle α; $p = k\sin\alpha$, where k is the wavenumber of waves in free-space [16]. Equation (48) and (49) describe continuous Fourier transforms, however, to implement them as a numerical method, discrete transformations must be made. Thus p is band limited such as the limits of integration $\pm\infty \rightarrow \pm p_{max}$ for discrete transforms. As a direct consequence, this also limits the integration in z, following the Nyquist criteria [16]. Thus a carrier frequency of 10 GHz corresponds to wavelength $\lambda = 3$ cm and a maximum propagation of angle of 5° will give a step size of 17 cm. To avoid errors in calculations a filter function is applied to get rid of faulty reflection from the top boundary. The field above the maximum height of interest are multiplied by a filtering function [16]. The derivation of the SPE (40), was made by assuming a constant refractive index. In reality, the refractive index varies with both range and altitude above the earth's surface and the above derived SSFT-PE fails to accurately account for those effects. Therefore, the model is extended by only considering refractive index with height dependence, i.e it is homogenous in the paraxial direction ($n = n(z)$). This will be the starting point and will be extended such that the refractive index has small variation with respect to the range. This will be included since lateral effects can significantly affect signal attenuation and ducting characteristics

[15].

The idea is to include height dependent refractive index $n(z)$ by instead approximate the operator Q such that the exponent in equation (38) can be split into two exponential operators as follows [16]

$$u(x + \Delta x, z) = e^{\delta(A+B)} u(x, z) \tag{52}$$

where A, B and δ are defined as

$$A = \frac{1}{k^2} \frac{\partial^2}{\partial^2 z^2} \tag{53}$$

$$B = n^2(z) - 1 \tag{54}$$

$$\delta = \frac{ik\Delta x}{2}. \tag{55}$$

We can split the two operators and instead write them as a product of exponentials only containing A or B and denoting the split operator as S_1. The factors of A can clearly be expressed as Fourier transforms meanwhile the operator B just becomes an multiplication factor. Thus the split operator S_1 can be written as

$$S_1 = e^{\delta A} e^{\delta B}. \tag{56}$$

The splitting of these operator is only valid for a homogenous medium. Otherwise the refractive index would be a function of height z (or height and range x). The operators A and B would in that case not commute yielding a small error in the split. To examine the magnitude of the error we calculate for the commutator. The commutator of two operators is defined as

$$[A, B] = AB - BA. \tag{57}$$

Two operators are said to commute if $[A, B] = 0$. Replacing A and B with equations (53) and (54) gives

$$[A, B]\, u(x, z) = -\frac{1}{2k^2} \left(\frac{\partial^2 n(z)^2}{\partial z^2} u(x, z) + 2 \frac{\partial n(z)}{\partial z} \frac{\partial u(x, z)}{\partial z} \right) \tag{58}$$

Thus the splitting introduced an error E which is expressed as

$$E = e^{\delta B} e^{\delta A} - e^{\delta(A+B)}. \tag{59}$$

By expanding the exponentials in a Taylor series the error can be expressed in terms of Δx, where the dominant error is given as [16]

$$\epsilon = \frac{k^2}{8}(\Delta x)^2 \left[A, B\right]. \tag{60}$$

Thus the error depends both on the range step, that usually is in the order of 10^2 m and the refractive index as can be seen from equation (59). Using the split the split-step solution at $x + \Delta x$ is given as [16]

$$u(x + \Delta x, z) = e^{\delta B} \left\{ e^{\delta A} u(x, z) \right\}. \tag{61}$$

The split step solution offers a good physical interpretation as the field can be viewed as propagating through a series of lenses as illustrated in figure 4. The Initial field propagates through a slice of homogenous medium that is

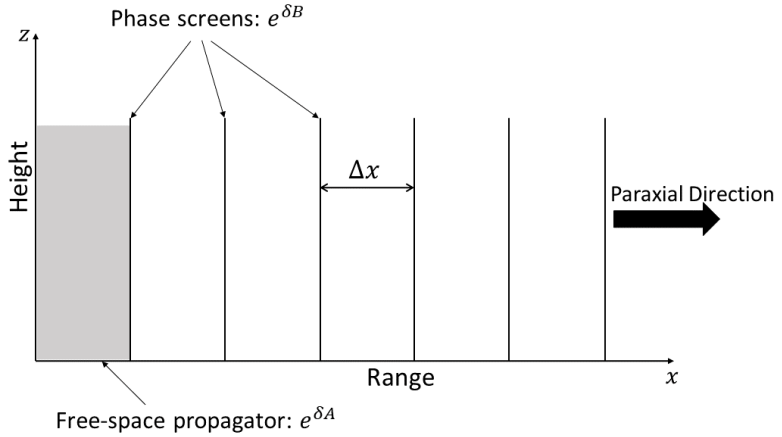


Figure 4: Schematic illustration of the phase-screen interpretation introduced by the splitting operator S_1

represented by the exponential $e^{\delta A}$. At the boundary, a phase screen modulation by refractive index variation is applied with the exponential $e^{\delta B}$. The split thus decouples diffractive and refractive effects which are embedded in $e^{\delta A}$ and $e^{\delta B}$ respectively and neglecting commutator effects. As previously mentioned before, the split introduces errors that can be minimized

by introducing another split operator. However, for radio-wave propagation applications, the refractive index usually have small variations and thus the error in the split is relative small. [16]

2.3.3 Range Dependence

Up till now, the refractive index has been approximated to only encapsulate variation with height. This is a fairly valid approximation when radio-waves propagate over land or over stable seas where the refractive index gradient remain sufficiently small over long ranges. However, in coastal areas where land meets sea, there is turbulence and land-sea air masses interchange causing range dependent refractive effects [15]. As previously mentioned the operator B encapsulates the refractive effects and to incorporate range dependent effects it is often rewritten as [16]

$$B_x = \frac{1}{\Delta x}\int_x^{x+\delta x} n(\xi, z)d\xi. \tag{62}$$

It is common to approximate the operator B as [16]

$$B = n(x + \frac{1}{2}\Delta x, z). \tag{63}$$

The effect on this can be seen as replacing the range dependent medium in figure 4 by a sequence of lenses with linearly varying refractive index. Similar to the range independent we would like to examine the magnitude of the error that the split operator infers. By using the simplest split S_1 defined by equation (56) the resulting error can be expressed as [16]

$$\begin{aligned} E &= e^{\delta A}e^{\delta B} - e^{\delta(A+B_x)} \\ &= e^{\delta A}e^{\delta B_x} - e^{\delta(A+B_x)} + e^{\delta A}(e^{\delta B-} - e^{\delta B_x}). \end{aligned} \tag{64}$$

By expanding the exponentials using a Taylor series and neglecting terms of order three or higher yield the following expression for the error [16]

$$\epsilon\frac{k^2}{8}(\Delta x)^2\,[A, B_x] \tag{65}$$

The expression for the commutator is similar to equation (58) but in this case the derivatives with respect to z is replaced by their average over a slice.

2.3.4 Wide angle Split Step Solution

The SSFT-SPE solution provided in previous section are accurate if the angles of propagation are close to the paraxial direction. This is normally the case when wave propagates in absence of terrain inclinations and unobstructed LOS. However, most of the time the terrain has large inclination and reflected waves might take of with larger angles than maximum angle α. Therefore, the approximation of the pseudo-vector Q must be re-evaluated to yield accurate results when waves propagates with angles larger than 10-15 degrees. The requirement for Q is that it shall be able to be combined with the split step formalism and without losing the computational advantage that the split step offers. Feit and Fleck proposed a operator split that since then has become popular when solving the SSFT-PE [31]. It is based on the approximation

$$\sqrt{1+a+b} \approx \sqrt{1+a} + \sqrt{1+b} - 1 \tag{66}$$

Using the previously derived definition of operator A, B and $u(x,z)$ given by equation (53), (54) and (30) and only considering forward propagation the psudo-operator can be approximated as

$$Q = \sqrt{1+A+B} \approx \sqrt{1+A} + \sqrt{1+B} - 1. \tag{67}$$

The error introduced by this approximation can be evaluated by taking the Fourier transform of the exact and approximated pseudo-operator given by [16]

$$\hat{Q}(p) = \sqrt{1+\delta n - \frac{4\pi^2 p^2}{k^2}} \tag{68}$$

$$\hat{Q}_1(p) = 1 + \delta n + \sqrt{1 - \frac{4\pi^2 p^2}{k^2}}. \tag{69}$$

Then by using Taylor expansion the error bound can be expresses as

$$\epsilon_1(p) = \left|\hat{Q}_1(p) - \hat{Q}(p)\right| \leq \frac{\pi^2 p^2}{k^2}\delta n \tag{70}$$

recalling that $2\pi p = k\sin(\alpha)$

$$\epsilon_1(p) \leq 4\sin^2(\alpha)\delta n \tag{71}$$

The same analysis for the narrow angle approximation yields an error proportional to $(\sin^2(\alpha) + \delta n)^2$. Thus the wide angle approximation is a great

improvement. The criterium when for introducing the wide angle propagator was that it have to be compatible with the SSFT-PE. This turns out to be the case. Replacing pseudo-operator in equation (35) with Q_1 yields the forward propagation wave function as

$$\frac{\partial u}{\partial x} = ik\left(\sqrt{1+\frac{1}{k^2}\frac{\partial^2}{\partial z^2}}-1\right)u + ik(n-1)u. \tag{72}$$

Similarly, the pseudo-operator in equation (38) gives the solution to the reduced function at $x+\Delta x$ as

$$u(x+\Delta x, z) = e^{ik\Delta x\left(\sqrt{1+\frac{1}{k^2}\frac{\partial^2}{\partial z^2}}-1\right)} e^{ik\Delta x(n-1)} u(x,z). \tag{73}$$

Employing the same technique that lead us to equation (51), the solution to the wide angle SSFT-SPE is given as [32]

$$u(x+\Delta x, z) = e^{ik(n^2-1)\Delta x}\mathcal{F}^{-1}\left\{U(x,p)e^{ik\Delta x\left(\sqrt{1+\frac{4\pi^2p^2}{k^2}}-1\right)}\right\}. \tag{74}$$

A comparison between the Narrow- and wide-angle propagators is presented in table 1. The wide angle SSFT-SPE will here on forth be used throughout this work. Since terrain is to modelled multiple reflections are diffraction occur yielding propagation angles larger than the $10^\circ - 15^\circ$ that the narrow angle are capable of accurately model. The wide-angle SSFT-SPE is therefore much better since it is capable of modelling propagation angles up to $40^\circ - 45^\circ$ [23].

Table 1: A comparison between the narrow- and wide-angle propagators.

Method	**Free space term**	**Environmental term**
Narrow angle:	$\exp\left(-\frac{4\pi^2 ip^2\Delta x}{2k}\right)$	$\exp\left(\frac{ik}{2}(n^2-1)\Delta x\right)$
Wide angle:	$\exp\left(ik\Delta x\left(\sqrt{1+\frac{4\pi^2p^2}{k^2}}-1\right)\right)$	$\exp\left(ik\Delta x(n-1)\right)$

2.4 Wave equations for propagation over the Earth

For waves to propagate b-LOS and reach a receiver it is required that the trajectory of the radio rays follow the curvature of earth. The SSFT-SPE formalism that has been developed has so far considered propagation over flat surfaces and in this section methods for incorporating the effects of spherical curvatures into the SSFT-SPE will be introduced. In the previous section wave propagation in two-dimensions were considered and we were able to express solutions to Maxwell's equation in terms of vertically and horizontally polarized fields. In general the solution does not split and instead a full three dimensional vector approach must be taken. In this section a derivation of Helmholtz wave equation using spherical coordinates will be provided. Further, for enabling solution with the use SSFT-SPE formalism the spherical wave equation must be transformed into a rectangular. It will be shown that the the curvature effects can be embedded into a modified refractive index $m(x, z)$.

2.4.1 Parabolic equation with Spherical coordinates

The earth is best represented by spherical coordinates the most suitable approach for tropospheric wave propagation would be using spherical coordinates (r, θ, ϕ) with the centre of earth as origin O. A schematic setup is illustrated in figure 5 where earth's radius is expressed as a, the transmitter (Source) is located on the z-axis and the point P denotes the location at which the field strengths is to be determined.
The atmosphere will be considered as a smoothly varying inhomogeneous medium with permeability μ corresponding to that of free space (μ_0). Similar to the 2-D case only harmonic time varying fields will be considered. Maxwell's equations then reads

$$\nabla \times \mathbf{E}(\mathbf{r}) = ikZ_0\mathbf{H} \tag{75}$$

$$Z_0\nabla \times \mathbf{H}(\mathbf{r}) = -ikn^2\mathbf{E}(\mathbf{r}) \tag{76}$$

$$\nabla \cdot n^2\mathbf{E}(\mathbf{r}) = 0 \tag{77}$$

$$\nabla \cdot \mathbf{H}(\mathbf{r}) = 0 \tag{78}$$

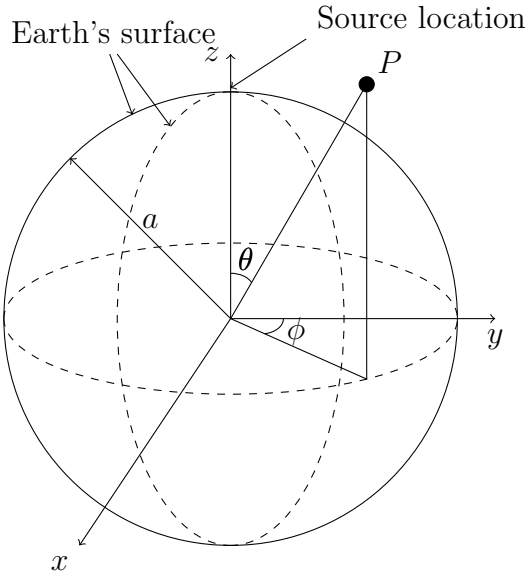


Figure 5: Illustration of the spherical coordinate system were the origin is set as the centre of the earth.

Where Z_0 is the impedance of free space ($Z_0 = 1/(\epsilon_0 c) \approx 120\pi$) for SI units. Taking the curl of equations (75) and (76) generates six scalar equations

$$\begin{cases} \frac{1}{r^2 \sin\theta}\left(\frac{\partial(rE_\phi \sin\theta)}{\partial\theta} - \frac{\partial(rE_\theta)}{\partial\phi}\right) & = ikZ_0 H_r \\ \frac{1}{r\sin\theta}\left(\frac{\partial E_r}{\partial\phi} - \frac{\partial(rE_\phi \sin\theta)}{\partial r}\right) & = ikZ_0 H_\theta \\ \frac{1}{r}\left(\frac{\partial(rE_\theta)}{\partial r} - \frac{\partial E_r}{\partial\theta}\right) & = ikZ_0 H_\phi \\ \frac{Z_0}{r^2 \sin\theta}\left(\frac{\partial(rH_\phi \sin\theta)}{\partial\theta} - \frac{\partial(rH_\theta)}{\partial\phi}\right) & = -ikn^2 E_r \\ \frac{Z_0}{r\sin\theta}\left(\frac{\partial H_r}{\partial\phi} - \frac{\partial(rH_\phi \sin\theta)}{\partial r}\right) & = -ikn^2 E_\theta \\ \frac{Z_0}{r}\left(\frac{\partial(rH_\theta)}{\partial r} - \frac{\partial H_r}{\partial\theta}\right) & = -ikn^2 E_\phi \end{cases} \tag{79}$$

Equation (79) is to complicated to solve numerically and need to be reduced. Generally, approximations based on symmetry properties, such as posing full angular- or azimuthal symmetry are used. The choice of approximation is determined by the application. Posing full angular symmetry would directly imply that the earth is a perfect sphere without terrain inclination and the atmosphere surrounding it is concentrically layered [16]. This would be far

too restrictive when looking at tropospheric propagation since it can not be any terrain in paraxial direction and hence this approximation is discarded in this study. Posing azimuthal symmetry is less restrictive and lets terrain profile and atmospheric conditions to vary along the paraxial direction while keeping them invariant in ϕ. This is in fact a quite good approximation since both terrain and atmospheric refractivity is slowly varying in the azimuthal direction [16]. Furthermore, since the wavelength of the waves are small compared to the radius of the earth most of the energy between transmitter and receiver is confined to great circle plane containing the antennas [16]. Assuming azimuthal symmetry does not imply that the fields themselves are invariant in ϕ, depending on the antenna pattern fields can have azimuthal dependence and thus the choice of antenna pattern can become crucial. It should also be noted that there are cases where lateral ground variations become important and give rise to multipath propagation. An example of that would be in urban environments with high building density. In those scenarios the full three dimension vector approach must be taken.

Equation (79) is simplified by assuming that field are invariant in ϕ and thus reduces to

$$\begin{cases} \frac{1}{r^2 \sin\theta}\left(\frac{\partial(rE_\phi \sin\theta)}{\partial\theta} - \right) & = ikZ_0 H_r \\ \frac{1}{r\sin\theta}\left(\frac{\partial(rE_\phi \sin\theta)}{\partial r}\right) & = -ikZ_0 H_\theta \\ \frac{1}{r}\left(\frac{\partial(rE_\theta)}{\partial r} - \frac{\partial E_r}{\partial\theta}\right) & = ikZ_0 H_\phi \\ \frac{Z_0}{r^2\sin\theta}\left(\frac{\partial(rH_\phi \sin\theta)}{\partial\theta}\right) & = -ikn^2 E_r \\ \frac{Z_0}{r\sin\theta}\left(\frac{\partial(rH_\phi \sin\theta)}{\partial r}\right) & = ikn^2 E_\theta \\ \frac{Z_0}{r}\left(\frac{\partial(rH_\theta)}{\partial r} - \frac{\partial H_r}{\partial\theta}\right) & = -ikn^2 E_\phi \end{cases} \tag{80}$$

From the boundary conditions given by (22) it is known that the tangential fields must be continuous at the ground/air interface. Due to azimuthal symmetry the tangential plane at any given point (r, θ, ϕ) is given by the unit vector $\mathbf{e}_\phi$ and the tangent vector $\boldsymbol{\tau}$ defined as

$$\boldsymbol{\tau} = \cos\alpha \mathbf{e_r} + \sin\alpha \mathbf{e}_\theta. \tag{81}$$

This implies that continuity of the fields tangential components is the same as continuity of E_ϕ, $\cos\alpha H_r + \sin\alpha H_\theta$ and H_ϕ, $\cos\alpha E_r + \sin\alpha E_\theta$ [16]. Therefore equation (80) can be split into two independent groups namely (H_r, H_θ, E_ϕ)

and (E_r, E_θ, H_ϕ). The two independent groups will now be used separately to further reduce the equations into a simpler form. The first case that will be studied is for $E_r = E_\theta = H_\phi = 0$ which corresponds to a vertical magnetic dipole source. The electric field only have one non zero component and by rearranging equation (80) and eliminating H_r and H_θ we arrive at the scalar wave equation

$$\frac{1}{r}\frac{\partial^2(rE_\phi)}{\partial r^2} + \frac{1}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial(\sin\theta E_\phi)}{\partial\theta}\right) + k^2n^2E_\phi. \tag{82}$$

Further simplifying the equation for E_ϕ by the substitution [16]:

$$E_\phi = \frac{1}{\sqrt{kr\sin\theta}}\psi_h. \tag{83}$$

Thus equation (82) in terms of ψ_h now reads:

$$\frac{\partial^2\psi_h}{\partial r^2} + \frac{1}{r}\frac{\partial\psi_h}{\partial r} + \frac{1}{r^2}\frac{\partial^2\psi_h}{\partial\theta^2} + \left(k^2n^2 - \frac{3}{4r^2\sin^2\theta}\right)\psi_h = 0 \tag{84}$$

Inferring the coordinate change

$$\begin{cases} X &= r\sin\theta \\ Z &= r\cos\theta \end{cases} \tag{85}$$

and rewriting equation (84) in terms of the new coordinate yields:

$$\nabla^2\psi_h(X,Z) + \left(k^2n^2 - \frac{3}{4X^2}\right)\psi_h(X,Z) = 0. \tag{86}$$

Due the earths large radius the term $1/X^2$ can be neglected and the result is a familiar wave equation in terms of ψ_h

$$\nabla^2\psi_h(X,Z) + k^2n^2\psi_h(X,Z) = 0 \tag{87}$$

For a vertical electric dipole source (i.e $H_r = H_\theta = E_\phi = 0$) the derivation becomes more complex. The only non-zero component of the magnetic field is H_ϕ and by eliminating E_r and E_θ the coupled equation (80) can be written as:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(\frac{1}{n^2}\frac{\partial(rH_\phi)}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{r^2\sin\theta}\frac{\partial(\sin\theta H_\phi)}{\partial\theta}\right) + k^2H_\phi = 0. \tag{88}$$

As in previous case we infer the substitution [16]:

$$H_\phi = \frac{n}{\sqrt{kr\sin\theta}}\psi_v. \tag{89}$$

In terms of ψ_v the wave equation reads as follows

$$\frac{\partial^2\psi_v}{\partial r^2} + \frac{1}{r}\frac{\partial\psi_v}{\partial r} + \frac{1}{r^2}\frac{\partial^2\psi_v}{\partial\theta^2} + (k^2n^2 + \kappa)\psi_v = 0. \tag{90}$$

The operator κ for a a stratified atmosphere is given by [16]:

$$\kappa = \sin\theta\left\{\frac{\partial^2\log n}{\partial r^2} - \left(\frac{\partial\log n}{\partial r}\right)^2\right\} + \frac{3}{4r^2\sin^2\theta} \tag{91}$$

For the case of a horizontally polarized source it was sufficient infer a change of coordinates and neglect the term $1/X^2$ to reach to the Helmholtz wave equation. This is not the case for a vertically polarized source due to the term κ. However, κ has two terms with derivates of $\log n$ with respect to r. As previously mentioned, variations of the refractive index is very small compared to the wavelength of the radiation. Therefore, the logarithmic terms in equation (91) can be neglected. By inferring the change of coordinates given by equation (85) the remaining term in equation (91) can be neglected since that corresponds to $1/X^2$ and again, we arrive at the familiar wave equation which in terms of ψ_v reads:

$$\nabla^2\psi_v(X,Z) + k^2n^2\psi_v(X,Z) = 0. \tag{92}$$

The polarization of the antenna will ultimately decide if equation (92) or equation (87) should be used to solve for ψ_v or ψ_h respectively.

2.4.2 Transformation to flat earth

In Section 2.3.2 the SSFT-SPE was derived by using a Cartesian coordinate system and in previous section we managed to transform the scalar wave equation to a rectangular coordinate system. Regardless of that the domain of interest is still better represented by spherical coordinates and therefore it is necessary to perform a transformation to instead obtain domain that could be expressed in a rectangular frame as illustrated in figure 6. There are multiple choices of transformation but in our case we are looking for a

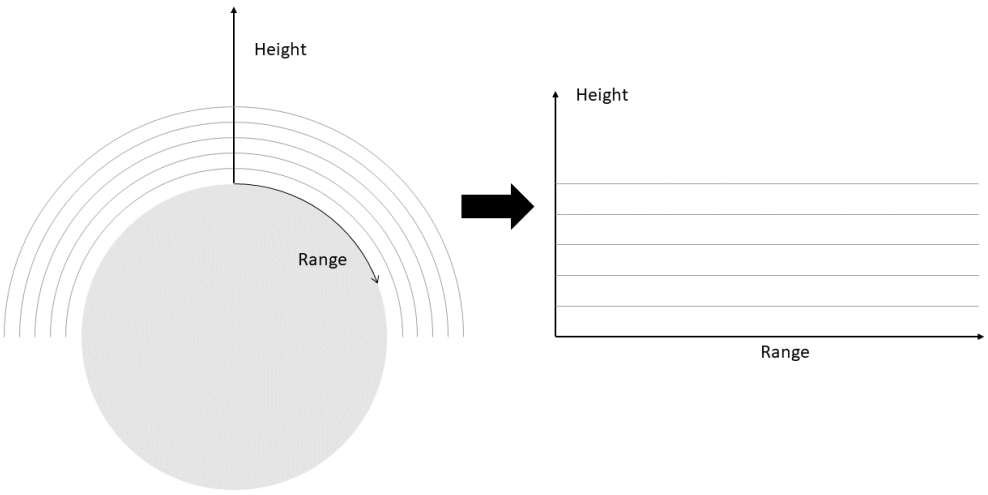


Figure 6: The flat earth transform

transformation that does not distort the wave equation and preserves local angles. A transformation that accomplishes this is one of the form [16]:

$$z = a \log\left(1 + \frac{h}{a}\right) \tag{93}$$

Where a is the radius of the earth. A point above earth surface can easily be expressed in terms of h by looking at the coordinate system illustrated in figure 5. We can place an antenna with height h at the position $a + h, 0, \phi$. Thus in our new coordinate system we would like the origin to coincide with the point $(a, 0, \phi)$ and for height to be measured in h and range in $a\theta$. The coordinate transformation in equation (85) can be written in terms of a and h as follows

$$\begin{cases} X &= (a+h)\sin\theta \\ Z &= (a+h)\cos\theta \end{cases} \tag{94}$$

Furthermore we define a circle in the complex plane with radius $r = a + h$

$$\zeta = X + iZ = i(a+h)e^{-i\theta} \tag{95}$$

Now we define a transformation in the complex plane outside the non-positive imaginary axis by [16]:

$$\xi = ia \log\left(\frac{\zeta}{ia}\right) \tag{96}$$

where ξ is given as

$$\xi = x + iz \tag{97}$$

Thus the transformation in equation (96) can be thought of as a change of coordinates from $(X, Z) \rightarrow (x, z)$[16]. By inserting equation (95) in equation (96) and using the principal determination of logarithms the following relation can be derived

$$\xi = a\theta + ia \log\left(1 + \frac{h}{a}\right). \tag{98}$$

Comparing with equation (97), the first term must correspond to $x = a\theta$ and the second term $z = a \log\left(1 + \frac{h}{a}\right)$. The chosen transformation have mapped a circle with centre 0 and radius $a + h$ constricted to the upper half of the plane to a horizontal segment $-\pi a \leq x \leq \pi a$ with height above the plane expressed as $z = a \log\left(1 + \frac{h}{a}\right)$. To validate this transformation we look at the difference between z and h. For a range of 500 km the difference is less than 22 m at a height of 5 km [16]. Clearly this can be seen as great approximation considering the fact that we will only bee concerned with height of $100 - 200$ m above earth's surface. However, as stated in the beginning of this section, the transformation must also preserve the wave equation. The remaining part of this section will derive the wave equation in the new frame.

The wave equations (87) and (92) has a Laplacian that is described in the old frame. By differentiating and applying Cauchy-Riemann theorems the Laplacian in new frame could be expressed as [33]

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} = \left|\frac{d\zeta}{d\xi}\right|^2 \left(\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Z^2}\right). \tag{99}$$

Thus the Laplacian in the old frame is proportional to that of the new frame. Using equations (96) and (99) it is straight forward to show that:

$$\left|\frac{d\zeta}{d\xi}\right|^2 = e^{2z/a}. \tag{100}$$

The wave equations (87) and (92) in the new frame can then be expressed as

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\tilde{n}^2 e^{2z/a}\psi = 0 \tag{101}$$

To get rid of the exponential a modified refractive index $\tilde{m}$ is defined as [16]

$$\tilde{m}(x,z) = \tilde{n}(x,z)e^{z/a}. \tag{102}$$

Since $z \approx h$ for height's that is small compared to the radius of the earth equation (101) can be written in the form [16]:

$$\tilde{m}(x,z) = n(x,h)\left(1 + \frac{h}{a}\right) = m(x,h). \tag{103}$$

The wave equation in the new domain reads

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial h^2} + k^2 m^2(x,h) = 0. \tag{104}$$

Where ψ is depending of polarization and in the form [16]:

$$\begin{cases} \psi_h = \sqrt{ka\sin\left(\frac{z}{a}\right)}e^{z/2a}E_\phi & \text{Horizontal polarization} \\ \\ \psi_v = \frac{1}{n}\sqrt{ka\sin\left(\frac{z}{a}\right)}e^{z/2a}H_\phi & \text{Vertical polarization.} \end{cases} \tag{105}$$

The chosen transformation accomplishes want we wanted. The wave equation is not distorted and the domain have successfully been transformed from spherical to rectangular.

2.4.3 Path Loss

In section 2.1 path loss was introduced and defined. Simulations results are often expressed in terms of path loss and we would therefore like to express the path loss in terms of the reduced field $u(x,z)$. The following derivation follows that of Levy in [16] where some steps have been omitted. For a more thorough derivation the reader is referred to [16]. For PE methods the transmission losses is expressed using a intermediate concept of path loss. Assuming that there are no system losses the transmission loss L_p is defined as

$$L_p = \frac{P_{iso}}{P_r(X,Z)} \tag{106}$$

where P_{iso} is the boresight equivalent isotropic radiated power for the antenna and P_r is the power received by an isotropic antenna with the same polarization [16]. The boresight equivalent isotropic radiated power is given by

$$P_{iso} = \frac{1}{Z_0} \tag{107}$$

where Z_0 is the impedance of vacuum. By treating the receiver as a point source the received power is given by

$$P_r(X, Z) = \frac{\lambda^2}{4\pi} S \tag{108}$$

where lambda is the wavelength of the EM-waves and S is the power flux density given by

$$\frac{1}{2Z_0} \left| E_\phi(X, Z) \right|^2 . \tag{109}$$

Inserting equations (107) and (108) into equation (106) and subituting $E_\phi(X, Z)$ with equation (83) gives

$$L_p = \frac{(4\pi)^2 X}{\lambda^3} \left| \psi_h(X, Z) \right|^{-2} . \tag{110}$$

By applying the flat earth transformation, the transmission loss in terms of the reduced field $u(x, z)$ is given as [16]

$$L_p = 20 \log_{10} \left(\frac{4\pi}{|u(x, z)|} \right) + 10 \log_{10} \left(a \sin \left(\frac{x}{z} \right) \right) - 30 \log_{10}(\lambda) \tag{111}$$

where a is te radius of the earth. Since $a >> x$ we can approximate L_p as

$$L_p = 20 \log_{10} \left(\frac{4\pi}{|u(x, z)|} \right) + 10 \log_{10}(x) - 30 \log_{10}(\lambda). \tag{112}$$

The derived formula for transmission loss hold for both vertically and horizontally polarized fields [16]. From here on forth transmission loss will be referred to as path loss.

Usually, results from radio-wave propagation simulation are often expressed using the propagation factor F and not the path loss. The propagation factor is defined as the field relative to that of free space and is given by [23]

$$F(x, z) = 20 \log_{10} \left(|u(x, z)| \right) + 10 \log_{10}(x) + 10 \log_{10}(\lambda) \tag{113}$$

where in [16] a misprint has unfortunately slipped in resulting in minus sign on the last two terms.

2.5 Boundary conditions

In section 2.2.1 general boundary conditions were discussed without mentioning the mediums or interfaces at all. To have a thorough discussion about boundary conditions the special domain boundaries must be addressed. The domain of interest is semi-infinite where the bottom boundary is bounded by the ground but supposedly infinite in upwards direction. In this chapter the ground boundary condition, domain truncation and initial conditions will be presented. The aim is to incorporate the presented boundary conditions in the SSFT-SPE formalism.

2.5.1 Ground

The general boundary conditions imposed by Maxwell's equations were presented in section 2.2.1. We are now ready to specify the interface and thus develop the necessary boundary conditions for solving the SSFT-SPE. We start by considering the interface between ground and air. The term ground is a bit loose but will be used to describe soil, water, asphalt or mountains. EM-waves from an antenna start by propagating in the air and after some range they come in contact with the ground. As radiation hits the ground parts of the EM-field is transmitted into the ground meanwhile some parts are refracted. The transmitted fields will not be considered in this thesis since they attenuate quickly inside the ground. Thus it is the the reflections that will be considered. The SSFT-SPE framework that so far has been presented only deals with either an electric field $\mathbf{E}$ or a magnetic field $\mathbf{H}$. Therefore, only two boundary conditions must be satisfied for any given case. For some application the ground can be approximated as a perfect electrical conductor and in that case either a Dirichlet (for horizontal polarization) or Neumann (for vertical polarization) boundary condition is sufficient. There boundary conditions are used when the ground is modelled as a perfect electrical conductor (PEC) and can be used for some scenarios. However, most problem are far more complex and require one to reduce the two conditions to just one. If the skin depth of the ground is small compared to radius of the earth, the tangential field continuity condition can be replaced by Leontovich boundary condition of the form [34][35]:

$$\left.\frac{\partial u(x,z)}{\partial z}\right|_{z=0} + \alpha u(x, z=0) = 0 \tag{114}$$

The skin depth is defined as the vertical length that fields propagate before its amplitude have decayed with a factor $1/e$ [34]. The skin depths for sea water at 20° and salinity 35 g/kg, dry and wet soil with volumetric water content 0.07 and 0.5 respectively are all below 1 m in the frequency range $1 - 10$ GHz[36]. Thus is reasonable to use a Leontovich boundary condition since the radius of the Earth usually is set as 6378 km .

Using the Fresnel reflection coefficients, Γ calculated in 2.2.1, the surface impedance α in the above equation can be calculated through [37]

$$\alpha = ik\cos(\theta_i)\left(\frac{1-\Gamma}{1+\Gamma}\right) \tag{115}$$

From Fresnel's equations developed in section 2.2.1 it is clear that (26) and (27) correspond to vertical polarization and horizontally polarized waves respectively. Thus inserting the Fresnel reflection coefficients into equation (115) and simplifying we end up with the following relation for α [37]

$$\alpha = \begin{cases} ik\frac{\sqrt{\epsilon_r - \sin\theta_i}}{\epsilon_r} & \text{For vertical polarization} \\ \\ ik\sqrt{\epsilon_r - \sin\theta_i} & \text{For horizotal polarization.} \end{cases} \tag{116}$$

Where θ_i is the incident angle measured from the surface normal and the complex relative permittivity of the earth surface is of the form [37]:

$$\epsilon_r = \frac{\epsilon_s}{\epsilon_0} + i\frac{\sigma}{\omega\epsilon_0}. \tag{117}$$

Where ϵ_s and σ is the absolute permittivity and conductivity of the surface respectively. It is possible to approximate the incident angle as 90° and therefore the sine terms in equation (115) could be set to a constant value of 1. This approximation is valid since the propagation range is far longer than the height of the antenna thus yielding large incident angles or small grazing angles. The values og the complex relative permittivity that will be used in this thesis are the ones specified in the ITU specification report [36]. In the following section the Leontovich boundary conditions will be included in the SSFT-SPE model.

2.5.2 Discrete Mixed Fourier Transform

In this section a method for incorporating the impedance boundary conditions into the SSFT-SPE formalism will be introduced. The derivation will follow that of Dockery and Kuttler [37]. The idea is to incorporate boundary effects by replacing the continuous Fourier transform with a discrete mixed Fourier transform (DMFT) consisting of sin and cos transforms. The impedance boundary condition is realised through a mixed Fourier transform of the form [37]:

$$U(x,p) = \int_0^\infty u(x,z)(\alpha \sin(pz) - p\cos(pz))dz. \tag{118}$$

The inverse transform is given by [38]

$$u(x,z) = \frac{2}{\pi}\int_0^\infty F(p)\frac{\alpha\sin(pz) - p\cos(pz)}{\alpha^2+p^2}dp + K(x)e^{-az} \tag{119}$$

where [38]

$$K(x) = \begin{cases} 2\alpha\int_0^\infty u(x,z)e^{-az}dz & \mathrm{Re}(\alpha) > 0 \\ \\ 0 & \mathrm{Re}(\alpha) \leq 0 \end{cases}. \tag{120}$$

The mixed Fourier transform split step solution is then given by substituting equation (119) into the wide angle PE given by equation (74)

$$\begin{aligned} u(x+\delta x, z) =& e^{i(k/2)(\tilde{m}-1)\Delta x} \\ & \left\{ \frac{2}{\pi}\mathcal{F}_s\left[\frac{\alpha}{\alpha^2+p^2}e^{i\Delta x(\sqrt{k^2-p^2}-k)}U(x,p)\right]\right. \\ & -\frac{2}{\pi}\mathcal{F}_c\left[\frac{p}{\alpha^2+p^2}e^{i\Delta x(\sqrt{k^2-p^2}-k)}U(x,p)\right] \\ & \left. +e^{i\Delta x(\sqrt{k^2-\alpha^2}-k)}e^{-\alpha z}K(x)\right\} \end{aligned} \tag{121}$$

Where $\mathcal{F}_s$ and $\mathcal{F}_c$ is the sine and cosine transform respectively defined as:

$$\begin{aligned} \mathcal{F}_s\left[f(x,z)\right] &\equiv \int_0^\infty f(x,z)\sin(pz)dz \\ \mathcal{F}_c\left[f(x,z)\right] &\equiv \int_0^\infty f(x,z)\cos(pz)dz \end{aligned} \tag{122}$$

and with

$$U(x,p) = \alpha \mathcal{F}_s \left[e^{i(k/2)(\tilde{m}-1)\delta x} u(x,z) \right] - p\mathcal{F}_c \left[e^{i(k/2)(\tilde{m}-1)\delta x} u(x,z) \right]. \quad (123)$$

This is usually referred to as the mixed Fourier transform and have been used to solve various problems where the surface can not ideally be approximated as perfect electrical conductors. However, numerical problems may arise when $\mathrm{Re}(\alpha)$ is very small and $\mathrm{Im}(\alpha)$ is smaller than p_{max} than the denominator tends to zero for some values of p, this problem can easily be solved by decreasing the mesh spacing which would correspond to increasing the transform size. Next we would like to find the discrete version of the mixed Fourier transform split step solution to be able to numerically handle it. The discrete mixed Fourier transform (DMFT) is [38]

$$F(j\Delta p) \equiv \sum_{m=0}^{N}{}' f(m\Delta z) \left[\alpha \sin\left(\frac{\pi jm}{N}\right) - \frac{\sin\left(\frac{\pi j}{N}\right)}{\Delta z} \cos\left(\frac{\pi jm}{N}\right) \right] \quad (124)$$

Where Δp is the sample point distance in transform space, Δz is the sample point distance in height in real space and $\Delta z \delta p = \frac{\pi}{N}$. The prime on the summation indicates that the first and last term is scaled with a factor of 0.5. To complete the transform the following quantities are needed [38]:

$$C_1 = D \sum_{m=0}^{N}{}' f(m\Delta z) r^m \quad (125)$$

$$C_2 = D \sum_{m=0}^{N}{}' f((N-m)\Delta z)(-r)^m \quad (126)$$

Where the quantity D is defined as

$$D = \frac{2(1-r^2)}{(1+r^2)(1-r^{2N})}. \quad (127)$$

The variable r is retrieved by solving for the roots of the quadratic equation

$$r^2 + 2r\alpha\Delta z - 1 = 0. \quad (128)$$

Since it is a quadratic equation two solution are attained and which one to use is determined by the antenna polarization.

$$r = \begin{cases} \sqrt{1+(\alpha\Delta z)^2} - \alpha\Delta z & \text{For vertical polarization} \\ \\ -\sqrt{1+(\alpha\Delta z)^2} & \text{For horizontal polarization} \end{cases} \quad (129)$$

With this parameters defined the full DMFT is given by [38]:

$$f(m\Delta z) = \frac{2}{N}\sum_{0}^{N}{}' F(j\delta p)\frac{\alpha \sin\left(\frac{\pi jm}{N}\right) - \frac{\sin\left(\frac{\pi j}{N}\right)}{\Delta z}\cos\left(\frac{\pi jm}{N}\right)}{\alpha^2 + \left[\frac{\sin\left(\frac{\pi j}{N}\right)}{\Delta z}\right]^2} + C_1 r^m + C_2(-r)^{(N-m)} \tag{130}$$

Even though the DMFT avoided a series of continuous Fourier transform by converting it to a discrete frame the computational efficiency remains small. Calculating the field at a range step $x + \Delta x$ still requires 2 transforms and thus it becomes time when dealing with large domains. The split step solution can be thought of as a step by step algorithm with utilizes the field derived from the point $u(x, m\delta z)$ to calculate the field at $u(x + \Delta x, \delta z)$. To decrease computation times Dockery and Kuttler also provided a method based on only discrete sine transforms [38], this was later improved by Kuttler and Janaswamy in [39] where they also dealt with some of the bad alpha problems that was previously mentioned. However, we will not go into detail of how the discrete sine transform work instead the interested reader is referred to [38] and [39].

2.5.3 Initial condition

The SSFT-PE is a marching algorithm and initial condition (i.e the reduced field $u(x_0, z)$ where x_0 is the range at which the source is placed) must be properly specified since this is our basis vector. The initial field is completely given by the source. In this project we will be following the work of Levy [16] to express the initial field by an Gaussian beam pattern. The initial field is given through

$$u_{fs}(0, z) = \frac{k\beta}{2\sqrt{2\pi \log(2)}} \exp(-ik\theta_0 z) \exp\left(-\frac{\beta^2}{8\log(2)} k^2 (z - h_a)^2\right). \tag{131}$$

Where θ_0 is the elevation angle of the antenna with the ground beneath the antenna being the reference point, β is the antenna half power beam width, h_a is the antenna height and k is the wave vector of the radiation. The position of the antenna has been chosen as $(0, z)$

The initial field must satisfy the boundary condition. Assuming that the ground beneath the antenna is perfectly conducting. the initial field is written as [16]:

$$u(x=0,z) = \begin{cases} u_{fs}(0,z) - u_{fs}(0,-z) & \text{Horizontal Polarization} \\ u_{fs}(0,z) + u_{fs}(0,-z) & \text{Vertical Polarization.} \end{cases} \tag{132}$$

The fields that are calculated are used as initial fields in DMFT-algorithm.

2.5.4 Domain Truncation

As mentioned at the start of this section the domain is semi-infinite and have no upper boundary. When performing simulation the domain must be cut off at some point. To avoid faulty results the waves that have reached the upper boundary can not come back and interact with EM-waves inside the domain. Intuitive it might feel like a good approach to just put the fields outside the domain to zero. However, this causes reflection due to the Fourier transforms. To avoid faulty reflections a window function also called absorption layer is added which slowly attenuate the field that reached through the top of the domain. For this work the Hann-window is chosen and defined as [16]

$$w(z) = 0.5\left(1 - \cos\left(2\pi\frac{z}{Z}\right)\right) \tag{133}$$

Where Z is the length of the window.

The Hann-window function is multiplied to each sampled field value in the absorption layer. The window-function is numerically inconvenient since the domain size increases but it is still widely used among scholars to get rid of faulty reflections. However, the optimum size of the absorbing layer is still not determined. When fields propagates at small angles i.e close to the paraxial direction the fields do not propagate far inside the absorption layer and its enough to have a 100 m window to properly attenuate the fields [16].

2.6 The Atmosphere

In network systems an antenna emits electromagnetic waves that propagate in the lower part of the atmosphere known as the troposphere to reach a specific receiver. The constituents of the atmosphere depends on the altitude

above earth surface but can in general be seen as quasi uniform distribution up to 15-20 km where the main constituents with percentage of the total volume are: Nitrogen (N_2), 78.095%; Oxygen (O_2), 20.93%, Argon (Ar), 0.93% and carbon dioxide (CO_2), 0.03% [40]. The refractive index of the atmosphere are dependent on the constituents and hence also the altitude. Electromagnetic waves that propagates in the atmosphere are thus refracted differently depending on the constituents of the air. The atmospheric refractive index can be divided into a real and complex part where the real part is responsible for effects such as refraction, ray bending, multipath, depolarisation and ducting [40]. The real part of the index of refraction for a given medium is defined as the ratio between the speed of light in vacuum and the speed of the light in that particular medium [26].

$$n = \frac{c}{v} \tag{134}$$

The complex part of the refractive index can be used to describe effects such as absorption and signal attenuation due to atmosospheric perturbation (Rain, snow and hail) [40].

In this thesis we will direct our focus to the real part of the refractive index and the term atmospheric refractive index will here forth be used to describe only the real part excluding the complex part. In this section the real part of the refractive index will be discussed. The reader will be introduced to concept of anomalous wave propagation which can enhance the range of radio wave propagation to beyond line of sight.

2.6.1 Refractive Index

The refractive index for air lies very close to unity and it rarely exceeds 1.0003, to avoid dealing with such small numbers and variation it is common to introduce the refractivity N defined as[41]:

$$N = (n - 1) \times 10^6. \tag{135}$$

The refractivity is a semi-empiricial given by the expression [40]

$$N = 77.6\frac{P}{T} - 5.6\frac{e}{T} + 3.75 \times 10^5 \frac{e}{T^2} \tag{136}$$

Where T is temperature in Kelvin, P is the atmospheric pressure in hectopascal (hPa) and e is the water vapour partial pressure in hectopascal

(hPa). Equation (136) has an accuracy of $\pm 0.5\%$ and by only considering temperature spanning from $-40^\circ - 40^\circ$ C it can be approximated as

$$N = 77.6\frac{P}{T} + 3.73 \times 10^5 \frac{e}{T^2} \tag{137}$$

The above equation is a standard formula recommended for use by the ITU in [42]. The atmospheric refractivity has a much greater vertical variation than horizontal and therefore is often assumed that a standard atmosphere only depends on the height above earth surface. It decreases with altitude as follows

$$N = 315e^{-\frac{z}{7.35}} \tag{138}$$

where z is the height above the ground in meters. As can be seen in figure 7 a wave propagating in standard atmosphere tends to bend downwards as a consequence of Snell's law derived in Section 2.2.1. Since the earth is curved and therefore it is of interest in knowing how EM-waves bends in relation to curved to the curvature of the earth. Most common way of including this effect is by introducing the modified refractivity M defined as [40]

$$M = 10^6 \times (m-1) = 10^6 \times \left(n - 1 + \frac{z}{a_e}\right) = N + 10^6 \times \frac{z}{a_e} \tag{139}$$

where z is the height above the local surface of the earth and a_e is the radius of the earth (6378 km). In the left part of figure 8 the modified refractivity as a function of altitude for a standard atmosphere can be seen.

What effects the EM-wave propagation is actually not the value of M rather it is the gradient that defines the propagation behaviour. The four categories of wave propagation are standard, ducting, sub-refractive and super-refractive. The values of gradients of M that causes these wave propagation mechanisms are shown in table 2 and in figure 7 an example of the ray trajectories in different atmospheric condition are illustrated where propagation in vacuum also have been included.

In sub-refractive conditions EM-waves will bend less than a standard atmosphere and can in some cases bend upwards. Since energy then spreads into space the signal strength is greatly decreased.

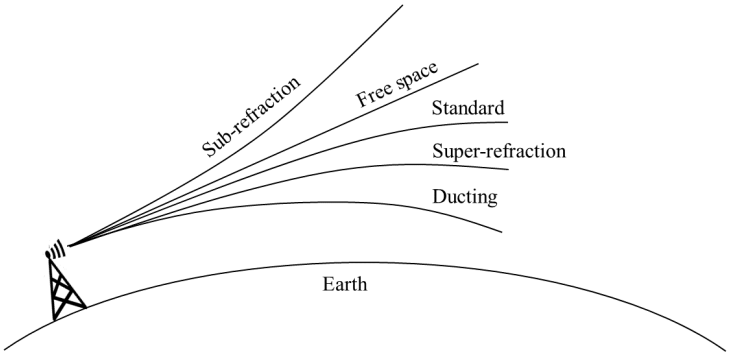


Figure 7: Illustration of the ray trajectory under different atmospheric refractivity conditions.

In Super-refractive conditions on the other hand, EM-waves are bent more than in standard atmosphere causing them to follow the curvature of the Earth longer. The energy is therefore more confined resulting in signal enhancements and increased range of the communication system.

It is not only the refractivity that affects how the EM-waves propagates in the troposphere. Precipitation can cause scattering and absorption and can result in decreased signal strengths or multipath propagation. Precipitation can for example be weather features such as rain, snow and hail and due to their small sizes are hard to incorporate in an model. Most often the effect of precipitation in the atmosphere are incorporated via a dampening factor.

Before moving forward with a more thorough discussion about atmospheric ducting conditions it is worth mentioning that so far only height variations of the refractive index have been considered. The atmosphere do actually change with range, however, those variations usually occur on large scales up to 10 of km and are dependent on the underlying terrain as well. This will be further discussed in the next part of this section.

Table 2: Atmospheric conditions and the corresponding modified refractive index gradient where the units are in M-units/km [1].

Atmospheric condition	$\mathbf{\frac{dM}{dz}}$
Sub-refraction	> 118
Standard Atmosphere	118
Super-refraction	$0 < \frac{dM}{dz} < 118$
Ducting	< 0

2.6.2 Ducts

Anomalous propagation or duct propagation is an extreme case of super-refractivity and occur when their is a region in the atmosphere that has a negative modified refractivity gradient. Waves in this layer can be seen to propagate in wave-guide like manner and could yield propagation far beyond the conventional radio line of sight [11]. A atmospheric duct can be either a surface based or an elevated duct as seen in middle and right part figure 8 respectively. EM-waves propagating in a surface based duct bounces between the surface of the earth and the top of ducting layer meanwhile for elevated duct it bounces between top and bottom of the duct. For incoming electromagnetic waves to be trapped inside a ducting layer their angle into the duct must be close to the paraxial direction or to the direction of the duct [6].

By looking at equation (137) it is clear that for the refractive index to have a large negative slope there most either be a sharp decrease of vapour pressure or a temperature inversion layer near the surface. Ducting layers are usually observed from the surface of the earth up to several kilometres in height. The thickness of the ducting layer can vary from being a few metres up to hundred of metres. Similarly they can extend hundred of kilometres in range or be local and extend just a few metres. To be able to use ducting as a long-range communication channel it is important that the duct extend

over large ranges and that they are nearly permanent in time. Over watered bodies (seas, oceans, lakes) water is almost constantly evaporated. Close to the surface there is a rapid decrease of moisture in the air as the altitude increases. This causes evaporations ducts to form [11]. Evaporation ducts have appearance probability of around 90% in equatorial areas [4]. Thus long-range communication are especially well suited for over-seas communications [2]. The main purpose of this thesis is to examine wave propagation over watered bodies and therefore only the theory regarding evaporation ducts will be presented.

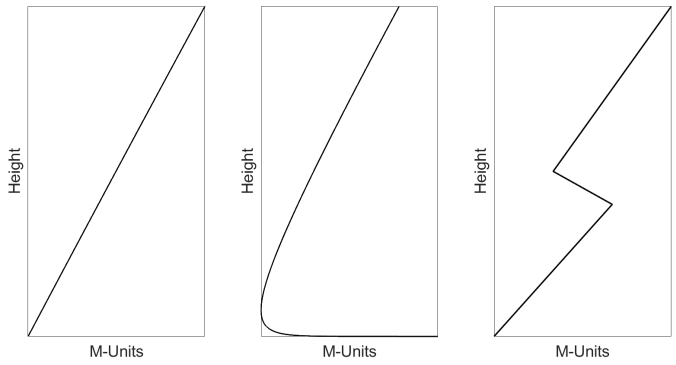


Figure 8: M-profiles as function of altitude. The left one is for standard atmosphere, in the middle an evaporation duct and to the right an elevated duct.

Even though evaporation ducts have a nearly constant presence in marine environments the strength and duct height a highly dependent on geographical location and season. Generally their height is less 40 m and their strength is between $5 - 20$ M-units strong. The duct is fairly constant over watered bodies. However, in coastal regions where more than one air mass interact a complex interplay occurs resulting variation of the M/z profile [15]. In [15] the author suggested that a range dependent duct could be possible to implement by linear interpolation between different M-profiles. As can be seen in figure 8 the evaporation duct is not a bilinear curve. There are many different ways of modelling an evaporation duct. Dinc and Akan modelled a

evaporation with a logarithmic function as of the form [7]:

$$M(z) = M_0 + 0.125 - 0.125\delta \log\left(\frac{z + z_0}{z_0}\right) \tag{140}$$

where M_0 is the modified refractivity at the surface, δ is the duct height and z_0 is the aerodynamic roughness lenght taken as 1.5×10^{-4} m. The Parabolic Equation Toolbox version 2 (PETOOLv2.0) developed by Ozgun et.al has gathered multiple evaporation ducts models into their MATLAB based toolbox [24]. Even though multiple models have been developed we will not go into detail explaining them. In this work we will instead approximate the evaporation duct as a bilinear surface duct. The main purpose is to develop a model that can take sea surfaces into account and range varying refractivity profiles. Once that is done a more rigorous model capable of handling different types of atmospheric ducts can be developed.

2.7 Terrain Modelling

PE methods have during the last decades become increasingly popular as method for radio wave propagation modelling. Before SSFT-SPE methods were introduced the geometrical optics(GO)/Geometrical Theory of Diffraction (GTD) methods was commonly used. Their advantage over PE-methods is that they are fast and able to include effects such as time delays. However, they rely on the atmosphere having slow changes and are not capable of handling irregular terrain features [16]. As electromagnetic waves hits the surface they are reflected and hence irregular terrain will ultimately lead to attenuations. The attenuation most be accounted for in order for the solutions to the SSPE-SPE to be accurate.

2.7.1 Ground

There are two common ways of implementing terrain features within the SSFT-SPE formalism. The staircase terrain modelling or the Piece wise linear shift map (PLSM).

The staircase method is simple and approximates the terrain as piecewise constant surface as illustrated in figure 9.

The idea with the staircase method is that as the field is marched forward in the horizontal segment S_1 the vertical boundary S_2 is ignored, where S_1

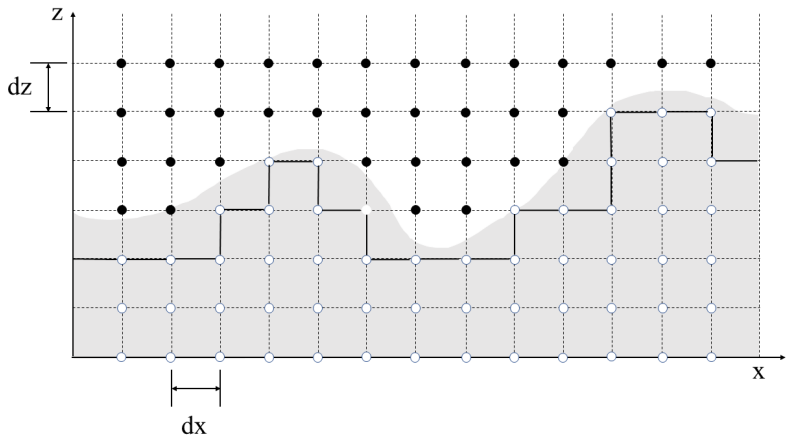


Figure 9: Terrain represented as a Staircase. Gray part is the real terrain and the black lines illustrates the Staircase terrain.

and S_2 can be seen in figure 10. When the field reaches the vertical boundary S_2 the terrain could have ascending or descending steps. For a ascending step the field is truncated at S_2 by setting it to zero. Thus ignoring backscattering. If the steps are descending the field is again truncated by setting it as zero on S_2, thus backscattering due to corner diffractions are neglected [16]. The idea behind the staircase method is simple and easily implemented in the SSFT-SPE model. However, the field strength tends to be overestimated unless the terrain is a perfect staircase .

The PLSM approach is a more complex method for handling irregular terrain features in which the terrain is modelled by piecewise linear segments. What is problematic about PLSM is that the coordinate system for the terrain is not rectangular. Instead height is measured from the terrain and to fit within the SSFT-SPE formalism it has to be transformed into a rectangular frame. This can be made by a terrain flattening transformation of the form [43]:

$$\begin{cases} \xi(x) = x \\ \zeta(z) = z - T(x) \end{cases} \tag{141}$$

Where $T(x)$ is the height of the Terrain.

For the new coordinate system and with $\psi(x, z)$=$\Phi(\xi, \zeta)$ Helmholtz equa-

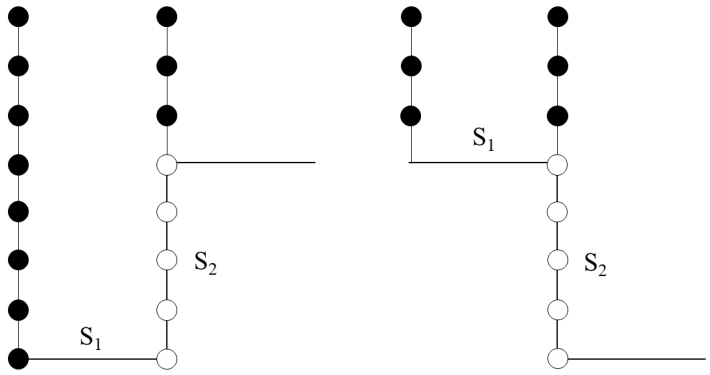


Figure 10: Illustration of the marching of the PE solution on a staircase terrain for a) Increasing terrain height b) Decreasing terrain height.

tion (Equation (30)) reads [43]

$$\left(\frac{\partial}{\partial x} - T'\frac{\partial}{\partial z}\right)^2 \psi(x,z) + \frac{\partial^2\psi(x,z)}{\partial z^2} + k^2 n^2 \psi(x,z) = 0 \tag{142}$$

Where $T' = dT/dx$.

For the above equation to be compatible within the split step formalism the first order derivate with respect to z must be eliminated. Defining a new function $\psi(x,z) = u(x,z)e^{ik\theta(x,z)}$, where $u(x,z)$ is the reduced wave function and $\theta(x,z)$ is a phase factor. Similar to when the SPE was derived the wave function can be separated into forward and backward propagating solutions. By making the substitution, the forward solution to equation (142) reads as follows [43]:

$$\left[\left\{\frac{\partial}{\partial x} + i\frac{\partial\theta}{\partial x} - T'\left(\frac{\partial}{\partial z} + i\frac{\partial\theta}{\partial z}\right)\right\} - i\sqrt{\left(\frac{\partial}{\partial z} + i\frac{\partial\theta}{\partial z}\right)^2 + k^2 n^2}\right] u = 0 \tag{143}$$

Continuing by choosing

$$\frac{\partial\theta}{\partial z} = k_0 T' \tag{144}$$

Which also implies that

$$\frac{\partial^2\theta}{\partial z^2} = 0 \quad \text{and} \quad \theta(x,z) = k_0 z T' + f(x)$$

with $f(x)$ being an arbitrary function and k_0 is a constant yet to be determined. With this choice of $\partial\theta/\partial z$ equation (143) reads:

$$\begin{aligned}\Bigg[\left\{\left(\frac{\partial}{\partial x} + if'\right) - T'\left(\frac{\partial}{\partial z} + ik_0T'\right)\right\} \\ -i\sqrt{\frac{\partial^2}{\partial z^2} + 2ik_0T'\frac{\partial}{\partial z} - k_0^2T'^2 + k^2n^2}\Bigg]\, u = 0\end{aligned} \tag{145}$$

The radical can be approximated as [43]:

$$\begin{aligned}\sqrt{\frac{\partial^2}{\partial z^2} + 2ik_0T'\frac{\partial}{\partial z} - k_0^2T'^2 + k^2n^2} &= K\sqrt{1+a+b+c} \\ &\approx K\left(\sqrt{1+a} + \sqrt{1+b} + \frac{1}{2}c - 1\right)\end{aligned} \tag{146}$$

where

$$K^2 = k^2 - k_0^2T'^2, \quad a = \frac{1}{K^2}\frac{\partial^2}{\partial z^2}, \quad b = \frac{k^2}{K^2}(n^2-1), \quad c = \frac{2ik_0T'}{K^2}\frac{\partial}{\partial z}.$$

With these definitions the radical can be expressed as:

$$\sqrt{K^2 + \frac{\partial^2}{\partial z^2}} + \sqrt{k^2n^2 - k_0^2T'^2} + \frac{ik_0T'}{K}\frac{\partial}{\partial z} - K. \tag{147}$$

Eliminating the term $\partial\theta/\partial x$ requires $k_0 = K \Rightarrow k_0 = \frac{k}{\sqrt{1+T'^2}}$. The function $f(x)$ can be arbitrary chosen and to obtain a simple expression for the piecewise linear shift map it is chosen such that [43]

$$f'(x) = k_0(-1 + T'^2). \tag{148}$$

Substitute the expression for x_0 and equation (148) into equation (145) gives [43]:

$$\frac{\partial u}{\partial x} = i\sqrt{\frac{k^2}{1+T'^2} + \frac{\partial^2}{\partial z^2}}u + ik\sqrt{\frac{n^2 - T'^2}{1+T'^2}}u \tag{149}$$

The equation above is known as the piecewise linear shift map. The slope of the terrain T' can also be written on the form $T' = \tan\beta$ where β is the angle that the terrain slopes makes with the horizontal. Thus k_0 can be written as $k_0 = \cos\beta$, meaning that the wave number is locally contracted. To put it in another perspective one can argue that tilting the coordinate system by an angle β locally contracts the range steps [43]. The piecewise linear shift map expressed by the angle β becomes:

$$\frac{\partial u}{\partial x} = i\sqrt{k^2\cos^2\beta + \frac{\partial^2}{\partial z^2}}u + ik\sqrt{n^2\cos^2\beta - \sin^2\beta}u \tag{150}$$

Since ϕ represents a physical it must be continuous at all points above the surface. However the slope of the terrain T' is not always continuous at the interface between two linear segment and hence the phase factor θ has a discontinuity. At the boundary between two linear segments the reduced function must therefore have a corresponding discontinuity given by [43]:

$$\begin{aligned} u_2(x_{1,2}, z) &= u_1(x_{1,2}, z)e^{ikz\left[(T_1'/1+T_1') - (T_2'/1+T_2')\right]} \\ &= u_1(x_{1,2}, z)e^{ikz[\sin\beta_1 - \sin\beta_2]}. \end{aligned} \tag{151}$$

The phase factor can therefore be seen as a steering factor that can be applied at every terrain slope discontinuity. The field is steered at the exact angle between the tilt of the two linear segments.

2.7.2 Ocean

The terrain on land can for the most part be accurately modelled since the height profile of the landscape can be measured and it rarely change over time. A sea surface on the other consist of waves which heights and wavelengths constantly change. Therefore, it is much harder to properly model a sea surface. How and when sea waves is generated does not lie in the scope of this thesis and therefore it will not be discussed. Instead we turn our attention to models that try to model the height of the surface and the reflections of the rough surface.

The effects of reflections from a rough sea surface can be incorporated in the Leontovich Boundary condition through a modified surface impedance of the form:

$$\alpha = ik\sin\theta\frac{1-\Gamma}{1+\Gamma} \tag{152}$$

where Γ is the effective reflection coefficient given as

$$\Gamma = \rho\Gamma_0 \tag{153}$$

where Γ_0 is the Fresnel reflection coefficient and ρ is the roughness reduction factor.

Assuming that the height $\zeta(x, z)$ of the surface is slowly varying compared to the wavelength of EM-waves, i.e the radius of curvature is far greater than the wavelength, the sea surface can be considered as locally flat. This is known as Kirchhoff's approximation and with this approximation the roughness reduction factor is given by

$$\rho = \int_{-\infty}^{\infty} \exp(2ik\zeta \sin\theta) P(\zeta) \tag{154}$$

where k is the wave number of the EM-waves, θ is the grazing angle and $P(\zeta)$ is a sea surface height probability density function. Miller et al. [44] proposed a probability density function of the form:

$$P(\zeta) = \frac{1}{\pi^{3/2}} \exp\left(\frac{\zeta^2}{8h^2}\right) K_0\left(\frac{\zeta^2}{8h^2}\right) \tag{155}$$

K_0 is the modified Bessel function of the second kind of order 0 and h is the root mean square height deviation given by the Philip spectrum as $h = 5.1 \times 10^{-3} U^2$ where U is the wind speed in m/s [16]. Using this expression for the PDF the roughness reduction factor is given by

$$\rho = \exp\left(-\frac{1}{2}\gamma^2\right) I_0\left(\frac{1}{2}\gamma^2\right) \tag{156}$$

where I_0 is the modified Bessel function of order 0 and γ is the Rayleigh roughness parameter given by

$$\gamma = 2kh \sin\theta. \tag{157}$$

Rayleigh's roughness parameter expresses the phase difference between two waves with grazing angles θ reflecting of two spots on the surface separated by h. The model proposed by Miller have extensively been used in the field of radio-wave propagation over rough sea surface. The drawback of Miller's

model is that is assumes all points on the surface as equally illuminated, thus is neglects shadowing effects. Furthermore, in the case of rough sea surface the surface impedance are dependent on the estimation of grazing angles and must be reconsidered at each step. Estimation of grazing angles can be made by two methods, either a spectral estimator or by Ray tracing (RT)/Geometrical optics (GO) methods. Spectral estimator such as the MUSIC algorithm [45] are powerful but tend to underestimate grazing angles when the atmosphere has strong refractive gradients [22]. Furthermore, spectral estimators in a DMFT algorithm first calculates the field at $u(x+\Delta x, z)$ with a Dirichlet boundary condition and then finds the dominant spectral component at the advanced step. The solution is then recalculated with surface impedance condition corresponding to the estimated grazing angle. Thus computational time is increased. Geometrical optics on the other hand is faster and able to handle evaporation ducts. However, GO methods are rather poor for estimating grazing angles in terrain shadows [22]. Furthermore, when ray-tracing is applied no rays will be detected at some ranges even though the field is non-zero [16]. For optimum grazing angles estimations a hybrid approach can also be used [22]

So far the geometry of the sea surface has not been considered. To Include the effects of sea surface geometry we follow the method presented by Benhmammouch et al. in [19] where the surface is considered to consist of large-scale roughness's and small-scale roughness's. To generate sea surfaces, sea surface spectrum are often used. The idea here is to divide the sea surface spectrum where one part is used to generate a sea surface corresponding to the large-scale roughness's and the second part is used to calculate the roughness reduction factor which then is incorporated into the Leontovich boundary condition through α. In this thesis we will be using the semi-empirical sea surface spectrum model proposed by Elfouihaily et al. [18].

The omnidirectional wave spectrum is expressed as [18]:

$$S(\kappa) = \kappa^{-3}(B_h + B_l) \tag{158}$$

where B_h and B_l are the curvature spectrum's for high and low frequencies respectively and κ is the angular spatial frequency in rad/m. The curvature spectrum for low frequency waves is given by [18]

$$B_l = \frac{1}{2}\alpha_p \frac{c_p}{c} F_p \tag{159}$$

where α_p is the Phillips and Kitaigorodskii equilibrium range parameters which depends on the wave age Ω_c [18]. In this thesis we assume that the sea is fully developed and therefore $\Omega_c = 0.84$ [18]. The parameter wave phase speed c given by

$$c = \sqrt{\left(\frac{g}{\kappa}\right)\left(1 + \frac{\kappa}{\kappa_m}\right)} \tag{160}$$

where g is the gravitational constant set to $g = 9.82$ ms^{-2} and $\kappa_m = 370$ rad/m [18]. The parameter c_p is the phase speed at the spectral peak k_p and is given by

$$c_p = \sqrt{\frac{g}{\kappa_p}} \tag{161}$$

where

$$\kappa_p = \kappa_0 \Omega_c^2 \quad \text{and} \quad \kappa_0 = \frac{g}{U_{10}^2}$$

where U_{10}^2 is the wind speed in m/s, 10 m above the surface. The function F_p is the long-wave side effect function given by [18]

$$F_p = LJ \exp\left(-\frac{\Omega}{\sqrt{10}}\left[\sqrt{\frac{\kappa}{\kappa_p}} - 1\right]\right). \tag{162}$$

The first and second term are given by [18]

$$L = \exp\left(-\frac{5}{2}\left(\frac{\kappa}{\kappa_p}\right)^2\right) \tag{163}$$

and

$$J = \gamma^{\Gamma} \tag{164}$$

where $\gamma = 1.7$ for fully developed sea and Γ is defined as

$$\Gamma = \exp\left(-\frac{\left(\sqrt{\frac{\kappa}{\kappa_p}} - 1\right)^2}{2\sigma^2}\right) \quad \text{and} \quad \sigma = 0.08[1 + 4\Omega_c^{-3}]. \tag{165}$$

The variable Ω is the inverse wave age given as

$$\Omega = \frac{U_{10}}{c_p}. \tag{166}$$

The curvature spectrum for high frequency waves is given by [18]

$$B_h = \frac{1}{2}\alpha_m \frac{c_m}{c} F_m \tag{167}$$

where c_m is the minimum phase speed at the wavenumber k_m given by

$$c_m = \sqrt{2g/\kappa_m} = 0.23 \text{ m/s}. \tag{168}$$

The short wave Phillips-Kitaigorodskii equilibrium range parameter is given by a two-regime logarithmic law as follows [18]

$$\alpha_m = 10^{-2} \begin{cases} 1 + \ln\left(\frac{u^*}{c_m}\right) & \text{for } u^* < c_m \\ 1 + 3\ln\left(\frac{u^*}{c_m}\right) & \text{for } u^* > c_m \end{cases} \tag{169}$$

where u^* is the friction velocity at the water surface given as [18]

$$u^* = \sqrt{Cd_{10N}} U_{10} \tag{170}$$

where the constant Cd_{10N} can be calculated from Figure 11 in [18] to correspond to $Cd_{10N} \approx 0.014$. The short-wave side effect function is takes as

$$F_m = LJ \exp\left(-\frac{1}{4}\left[\frac{\kappa}{\kappa_m} - 1\right]^2\right) \tag{171}$$

where in [18] there is a typo and the first term has got omitted.

In figure 11 the full Elfouhaily spectrum can be seen for wind speeds of 3-21 m/s with increments of 2 m/s.

The surface height is then generated from the Elfouhaily spectrum by using a Monte Carlo method. The surface height are given by [19]

$$h(x) = \sum_{i=1}^{N_s} \alpha_i \sin(\kappa_i x) + \beta_i \sin(\kappa_i x) \tag{172}$$

where α_i and β_i are stochastic variables drawn from a normal distribution with mean zero and variance $\sigma_i = 2\Delta\kappa S(\kappa_i)$ where $\delta\kappa$ is the step of discretization and $S(\kappa)$ is the Elfouhaily spectrum. Due to discretization there exist a maximum wave number κ_{max} that can be used to generate the sea

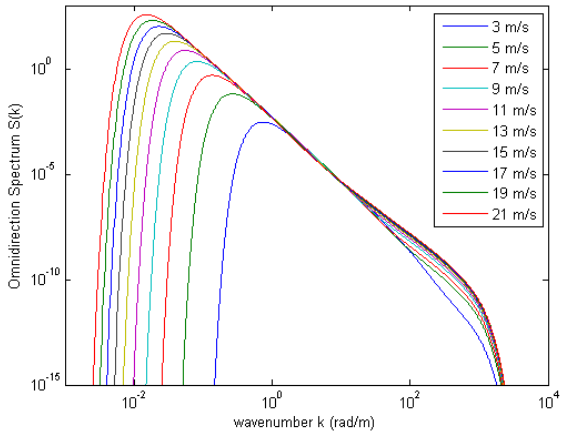


Figure 11: The calculated Elfouhaily ocean wave spectrum given the parameters defined in [18]. The different spectral curves corresponds to different winds speed ranging from $3 - 21$ m/s with increments of 2 m/s. The wave spectrum shown has a logarithmic scale.

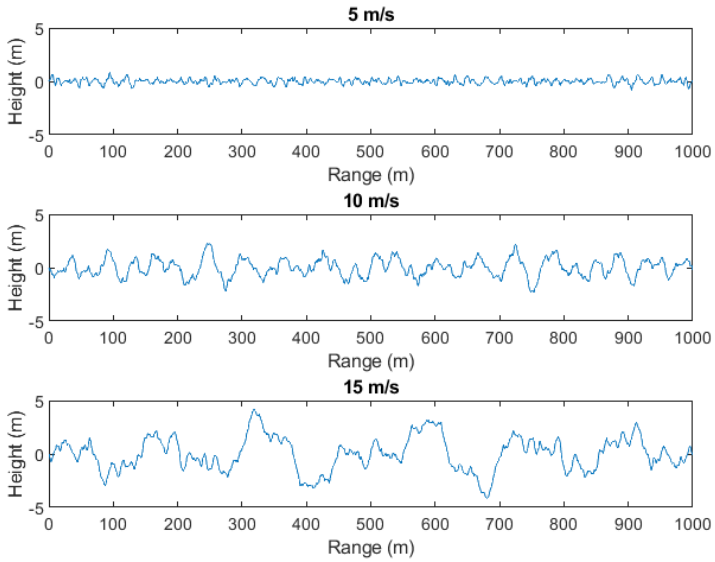


Figure 12: Sea surface generated by the method presented by Benhmammouch et.al in [19] and the Elfouhaily sea spectrum.

surface. All waves with wave number lower than κ_{max} can be used to generate the sea surface, and waves with wave number above κ_{max} will be considered in the modified impedance boundary condition. In figure 12, three generated sea surfaces for different wind speed can be seen.

To include the small-scale sea surface roughness in the modified sea surface impedance the root mean square height deviation h in equation (156) is replaced by the standard deviation of small-scale roughness h_s. The standard deviation of small-scale roughness is given by [19]

$$h_s = \sqrt{\int_{\kappa_{max}}^{\infty} S(\kappa) d\kappa}. \tag{173}$$

3 Method

This section aims to explain how the developed theory are implemented into an algorithm used for radio-wave simulations. The developed model can be split into two different categories, propagation over a perfect electric conductor (PEC) material and propagation over non-PEC material. In figure 13 an overview of the algorithm used for non-PEC material is shown in the form of a flowchart.

The algorithm will now be described in detail and related to the theory. The first part consist of specifying initial parameters such as antenna height, maximum range and height, mesh grid point, polarization etc. Next the type of terrain and atmospheric condition used for the simulation must be specified. Depending on the choice of terrain, height profile is either pre-specified or for a sea surface, calculated by using equation (172). The refractive index for each grid point is calculated using linear interpolation from a M vs z vector. The free space and environmental propagator is calculated through the equation for wide-angle propagators defined in table 1. The initial field that is used as starting field for the marching algorithm is calculated through equations (131) and (132).

The marching algorithm now start and is repeated for every step $x+N_x\Delta x$ where N_x is the number of grid points in range. The first step is to re-direct the field due to the terrain. This is made by multiplying the field with the steering factor defined in equation (151). The next steps are different depending on boundary conditions i.e is propagation over PEC material or not. If the ground is modelled as perfectly conducting then there is no need to determine the reflection coefficient and the marching algorithm is then the following: First transform the reduced field to the Fourier domain. Multiply the transformed field with the free space propagator. Perform the inverse Fourier transform and multiply with the environmental propagator to receive the reduced field u at $u(x + \Delta x)$.

For non-PEC materials the DMFT is instead used. The marching algorithm is again started by redirecting the field due to terrain. Next grazing angles is estimated with the help of a ray tracing program. The reflection coefficient is calculated either through equation (22) or equation (23), for a rough surface the reflection coefficient is multiplied with the roughness reduc-

tion factor in equation (156). For a flat rough surface the root mean square height deviation is calculated via the Philip spectrum as described in section 2.7.2. For a non-flat rough surface the standard deviation of surface wave height is calculated via equation (173). The reflection coefficient is then used to calculate the surface impedance α defined by equation (152). Next the parameter r is calculated via equation (129) which is subsequently used to calculate C_1, C_2 and D through equations (125)-(127). Next the DMFT is calculated via equation (124). The solution is then multiplied with the free space propagator and C_1, C_2 is propagated one step before the inverse DMFT is calculated through equation (130). The field is then multiplied with the environmental propagator to receive the field at the current step. Once the reduced field is calculated at each point the path loss and propagation factor is calculated through equation (112) and equation (113) respectively. The result is then presented either as a colormap showing propagation factor at each point or a plot of propagation factor vs range/height and compared to previous studies.

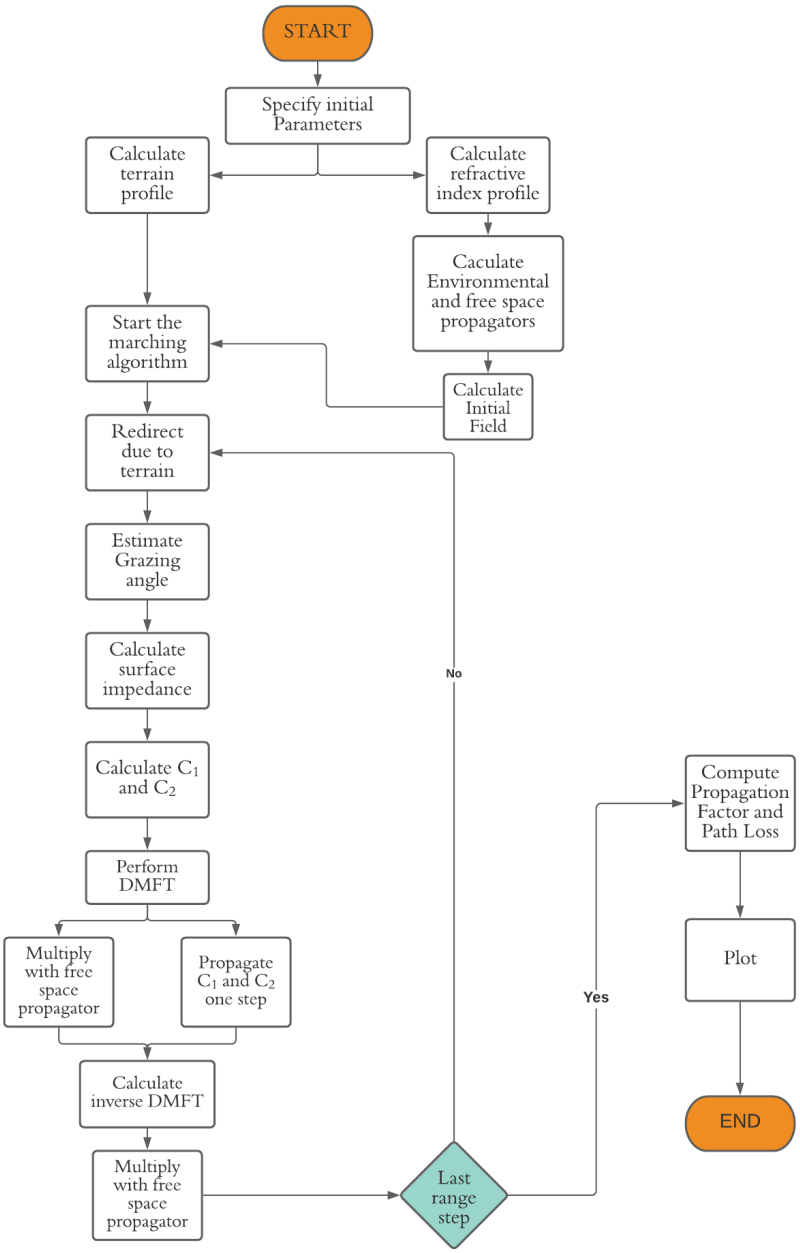


Figure 13: Flowchart for the algorithm used in this project.

4 Results and Discussion

In this section, the results obtained from the developed model are presented and discussed. The results are compared to similar studies to prove the accuracy of the model. Each subsection presents propagation simulations in different environmental conditions and terrains. The initial parameters that have been used to obtain the results can be found in Appendix A.

4.1 Propagation in Free space

This subsection will present results obtained for propagation in free space and standard atmospheric conditions. The ground has been modelled as a PEC for all simulation in this subsection. The results for free space simulation are compared results obtained using the free and publicly available MATLAB toolbox PETOOL v.2 Developed by Ozgun et.al in [24]. The results from propagation in standard atmospheric conditions are compared to PETOOL v.2 as well as figure 5 in [46], figure 3 in [47], and figure 7.5 in [16] to establish the accuracy of the model.

The results for free-space propagation simulations with a horizontally polarized antenna placed at an altitude of 50 m can be seen in figure 14. In figure 14a a colormap of the propagation factor is shown and in figure 14b the propagation factor at range 10 km is shown and compared to results obtained from PETOOL v.2. The first thing to notice is that the field at the surface equals zero. Since the surface is modelled as a PEC the boundary condition requires the field strength to be zero at the interface for horizontally polarized fields. Secondly, in figure 14b there is a visible lobe pattern caused by destructive and constructive interference between direct and ground reflected waves. The positive maximums are due to constructive interference and the negative peaks are due to destructive interference. All the maximums have similar value of propagation factor meanwhile the negative peaks have different depths. Since the mesh grid points are discrete the exact position of the minimums are probably not captured resulting in different depths. The agreement between the developed model and PETOOL v.2 is striking. The only visible difference is the depth of the minimums, otherwise the agreement is excellent.

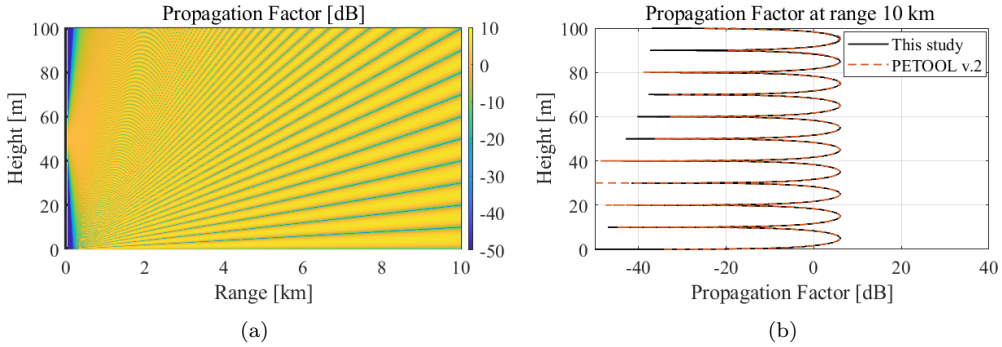


Figure 14: Wave propagation in Free space using a horizontally polarized field. a) Coverage diagram showing the propagation factor in free space using the developed model. b) Propagation factor as a function of height at a distance 10 km from the source for both the developed model and PETOOL v.2

The results for free-space propagation simulations with a vertically polarized antenna placed at an altitude of 50 m can be seen in figure 15. In figure 15a a colormap of the propagation factor is shown. In Figure 15b the propagation factor at range 10 km is plotted together with the results obtained from PETOOL v.2. The boundary conditions for a vertically polarized antenna is different and instead the derivative with respect to height must be zero to satisfy the boundary conditions. This is clearly seen in figure 15b where the first lobe maximum is located at the surface/air interface. The comparison between developed model and PETOOL v.2 again shows excellent agreement.

The next set of simulations concerns propagation in standard atmospheric conditions. The first simulation is performed with a horizontally polarized antenna placed at an altitude of 50m and the results are displayed in figure 16. In figure 16a a coverage diagram of the propagation factor is shown. The propagation factor at an altitude of 300 m for the developed model and PETOOL v.2 is plotted in figure 16b.

The most notable feature in figure 16a is that the field is bent upwards with respect to the surface. Beyond ≈ 40 km the propagation factor quickly attenuate close to the surface of the earth. This clearly illustrates the restricted radio LOS in standard atmospheric conditions approximation. The fact the rays bend upwards is a consequence of the flat earth transforma-

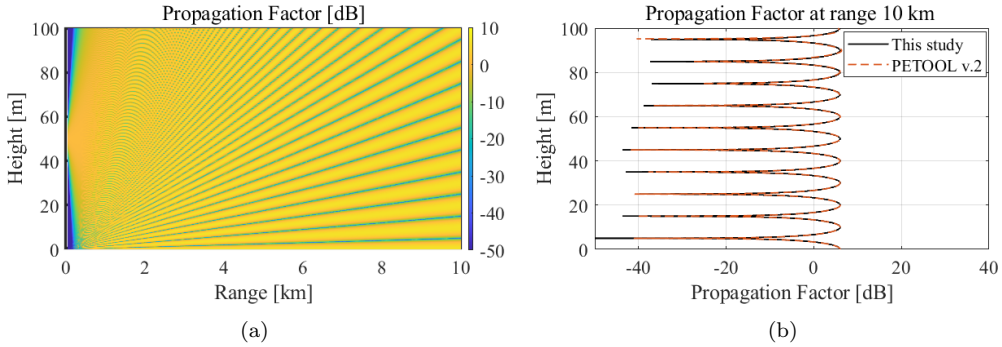


Figure 15: Wave propagation in Free space using a vertically polarized field. a) Coverage diagram showing the propagation factor in free space using the developed model. b) Propagation factor as a function of height at a distance 10 km from the source for both the developed model and PETOOL v.2

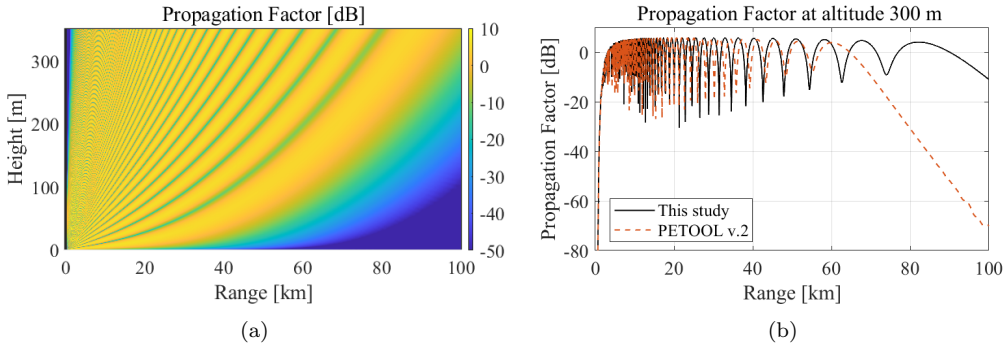
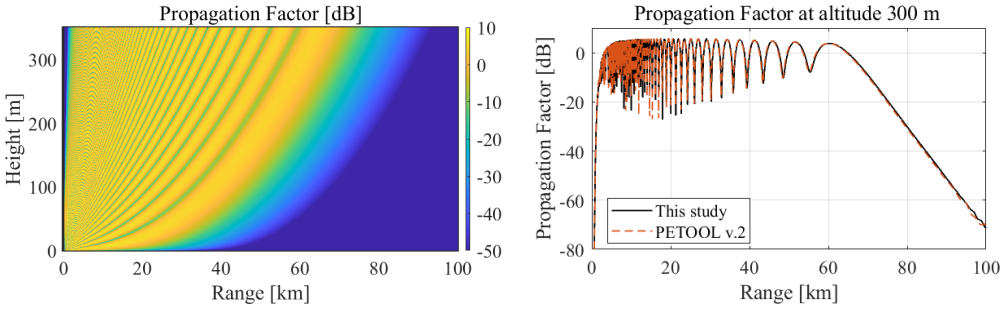


Figure 16: Propagation in standard atmosphere for a field with horizontal polarization. a) Color map obtained from the current model showing the propagation factor b) Propagation factor versus range at 300 m above the surface.

tion. If not applied, the surface would have been curved and the energy would have propagated with a radius of curvature larger than that of earth. Therefore, energy would have been propagating away from the surface of the earth. The result would be exactly the same but displayed in a different form.

As figure 16b shows, the agreement between the developed model and PETOOL v.2 is rather poor. In PETOOL v.2 the field are bent far more then in the developed model causing the propagation factor to attenuate at

much higher rate. However, the overall behaviour is similar and to find the difference between the models the environmental propagator in the developed model was modified such that $(m(x,z)-1) \rightarrow (m(x,z)^2-1)$. In figure 17 the modified environmental propagator have been used to obtain a coverage diagram of the propagation factor and the propagation factor at an altitude of 300 m.



(a) (b)

Figure 17: Propagation in standard atmosphere for a field with horizontal polarization. a) Color map obtained from the modified model showing the propagation factor b) Propagation factor versus range at 300 m above the surface.

By comparing figure 16a to figure 17a it is clearly seen that the modified environmental propagator causes stronger bending of the field. Using the modified environmental propagator the agreement between the developed model and PETOOL v.2 is outstanding as can be seen in figure 17b.

This raises some concerns about the developed model since both the non-modified and modified model can be simultaneously correct. Therefore, the non-modified version is compared to other sources as well. In the next simulation an horizontally polarized antenna is placed at an altitude of 31 m. The results are shown in figure 18 where (18a) shows a colormap of the propagation factor and (18b) shows the propagation at an altitude of 305 m.
The result presented in figure 18b is compared figure 5.5 in the work of Ehn [46]. Overall the agreement is great, with some minor differences. For example, the locations of the maximums are different. The maximums obtained by using the model presented in are placed roughly $0.5-1$km further forward than the ones in [46]. Furthermore, the minimums does not coincide either,

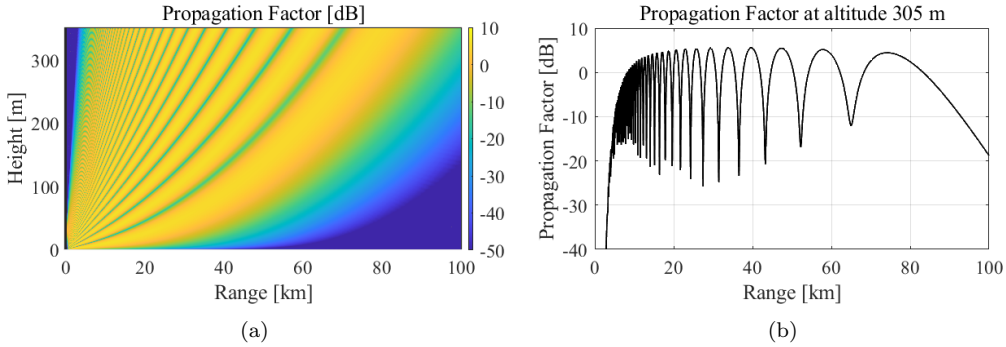


Figure 18: Propagation in standard atmosphere for a field with horizontal polarization. a) Color map obtained from the current model showing the propagation factor b) Propagation factor versus range at 305 m above the surface.

with the results from this thesis being lower than that of Ehn. The difference between the models could come from different modelling of the standard atmosphere, Ehn models the standard atmosphere with a refractivity gradient of $-42.84e^{-136z}$ N-units/km, whereas this study uses -39.2 N-units/km from the definition of refractive index gradient in [1]. Other than that the agreement is excellent.

To further establish the accuracy of the model another simulation is performed and compared to the path loss curve in figure 7.7 in [16]. The results from the simulation is shown in figure 19.
The only noticeable difference between figure 19b and figure 7.5 in [16] is the peak value at ≈ 5 km which for developed model is somewhat higher. This is probably due to the fact that smaller height and range increment was used compared to [16]. Furthermore, it is unclear how Levy modelled the atmosphere. Despite those small differences the overall agreement is very good and the results shows the similar behaviour.

The results from propagation in standard atmosphere with a vertical polarized antenna is displayed in figure 20. In figure 20a a coverage diagram can be seen and in figure 20b the propagation at 305 m above the surface is shown and compared to results from PETOOL v.2.

Similar to the case of a horizontally polarized antenna the rays are bent

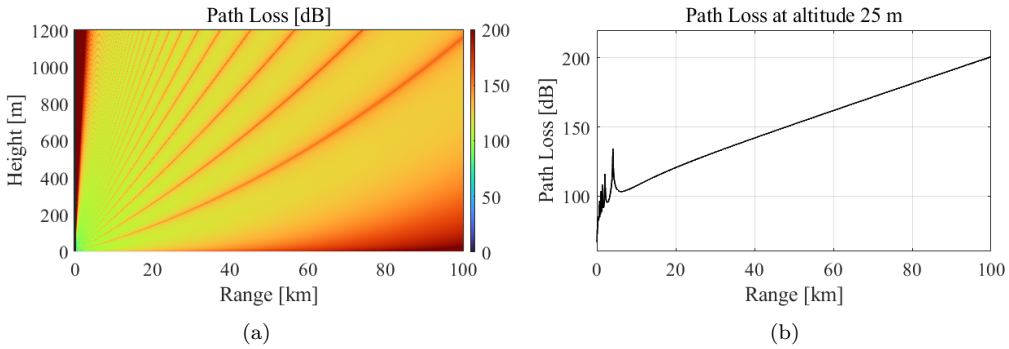


Figure 19: Propagation in standard atmosphere for a field with horizontal polarization. a) Color map obtained from the current model showing the path loss. b) Path loss as function of range, 25 m above the surface.

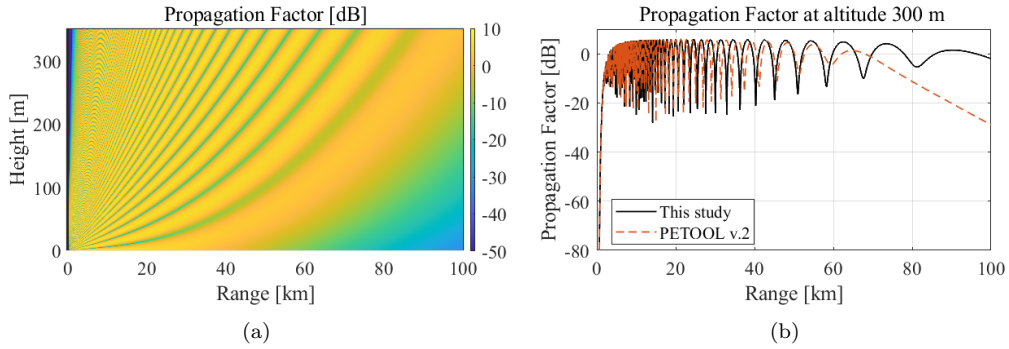


Figure 20: Propagation in standard atmosphere for a field with vertical polarization. a) Color map obtained from the current model showing the propagation factor b) Propagation factor versus range at 300 m above the surface.

upwards with respect to the surface of the earth. However, the propagation factor does not attenuate as quickly as in the case of a horizontally polarized antenna. The reason for this is the boundary condition and the fact that the field strength is non zero at surface. The agreement with PETOOL v.2 is rather poor but it again show the same general behaviour of the propagation factor. The difference in results are again due to modified refractive index. The results obtained by using the modified environmental propagator are shown in figure 21 where (21a) shows a coverage diagram of the propagation factor and (21b) shows the propagation factor at an altitude of 300 m.

The results shows an excellent agreement between the developed model and PETOOL v.2. The case of an vertically polarized antenna is not tested against any other sources. The reason for this is that in almost all literature, horizontally polarized antennas is used. Secondly, the ones that do use a vertically polarized antenna have instead chosen a different beam pattern which makes a qualitatively comparison hard. Lastly, the modified model agrees well with PETOOL v.2 for both types of polarization and since the non-modified model with a horizontally polarized antenna is verified through comparison to multiple sources, it feels safe to say that the developed model are accurate for both a horizontally and a vertically polarized source.

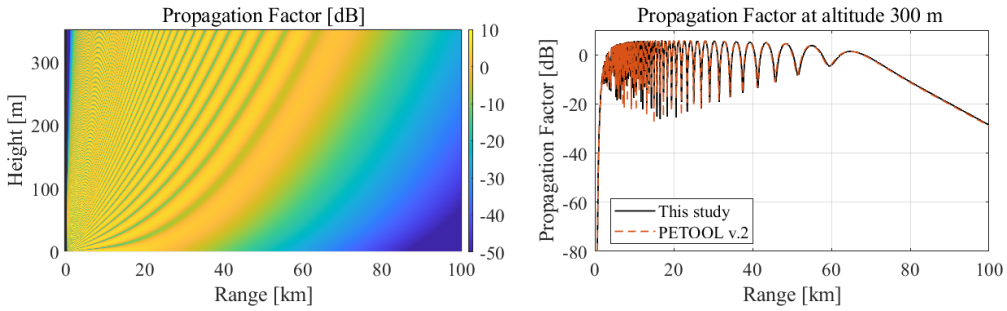


Figure 21: Propagation in standard atmosphere for a field with vertical polarization. a) Color map obtained from the modified model showing the propagation factor b) Propagation factor versus range at 300 m above the surface.

Even though the modified model showed excellent agreement to PETOOL v.2 the non-modified version could be compared to multiple sources that contradicted the results obtained from the modified model. Therefore the non-modified model will be used for the rest of this thesis. Comparison between the developed model to multiple sources showed good agreement. Therefore the model can be deemed as accurate for predictions of propagation losses in free space and standard atmospheric conditions.

4.2 Ducting

This subsection deals with wave propagation in surface ducts. The result will be presented as path loss colormaps instead of propagation factor since

path loss is often used for radio link budget calculations. In the first part a range independent surface duct will be used to study the the behaviour of electromagnetic waves propagating inside a ducting layer. The duct height was put as 20 m and the duct strength was put as 20 M-units. The layer stretching from the surface to the top of the duct are referred to as the ducting layer. Outside the ducting layer standard atmospheric conditions apply. The results will compared to figure 3.8 in [25]. The second part concerns surface ducts which strength and height are smoothly varying with range, the surface duct height and strength at each range can be found in Appendix A. All of the results presented in this chapter uses a horizontally polarized antenna and the ground is modelled as a PEC.

4.2.1 Range Independent

In figure 22, a coverage diagram of the path loss for waves with frequencies, (22a) 1 GHz, (22b) 5 GHz, (22c) 10 GHz, and (22d) 15 GHz propagating in the atmospheric conditions described above is shown.

First thing to be noticed is that the energy propagate in close proximity to the surface of the earth far beyond the radio LOS. The waves gets trapped inside the ducting layer which results in a decrease of path loss beyond the LOS compared to propagation in standard atmosphere. Secondly, not all parts of the field is confined within the ducting layer. Waves with large angle of arrival into the duct will not get trapped. Instead, they will propagate in a similar manner as waves in standard atmosphere. This is most noticeable for ranges up 40 km where similar behaviour as in figure 16 can be seen. This is one of the reasons why 5G network systems are promising for long range communication links. The antennas are highly directive which means that they can be directed in such way that most of the energy gets trapped inside the ducting layer.

As the frequency increases the behaviour of the propagating field drastically changes. The increased frequency causes stronger oscillations which in turn yields more multipath effects and interference lobes as can be seen in figure 22.

Figure 23a shows the path loss at an altitude of 25 m across the entire domain. Figure 23b displays the path loss as a function of height 200km from

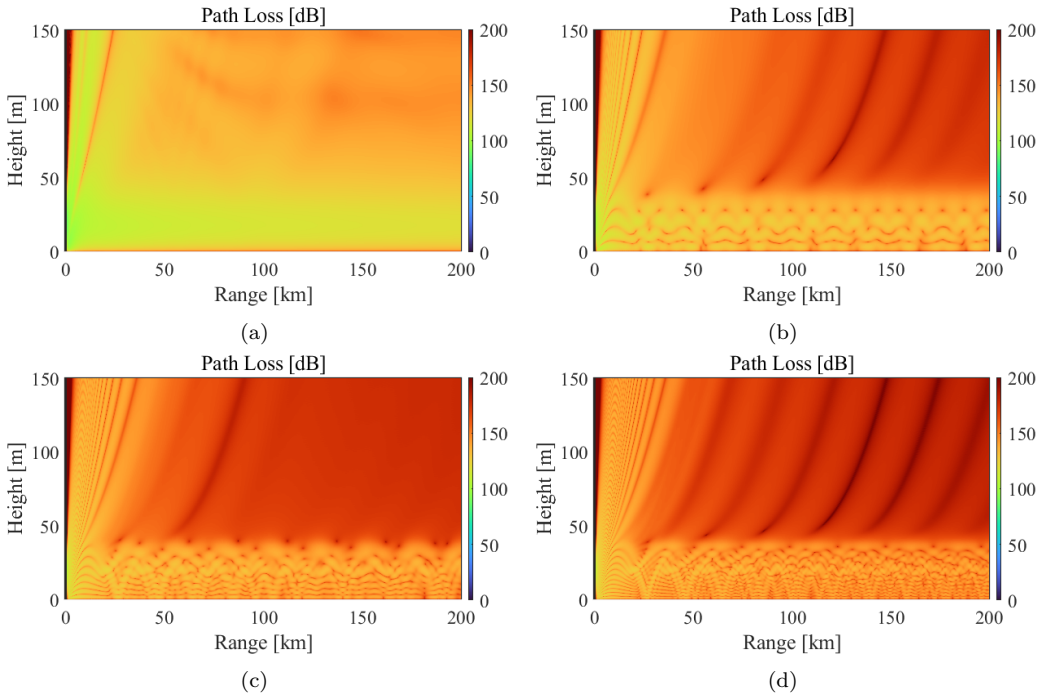


Figure 22: Path loss color maps for wave propagation in a bilinear surface duct described by table A2. The propagating waves have frequency a) 1 GHz, b) 5 GHz, c) 10 GHz and d) 15 GHz.

the antenna. For high frequencies the difference between the path loss inside the ducting layer and outside is larger, thus the trapping effects gets stronger as frequency increases. Secondly, notice that the drastic difference between path loss curves for 1 GHz and 5GHz waves. As soon as multipath becomes visible the path loss drastically increases. Further increasing the frequency and therefore the multipath effects does not yield a drastic difference in path loss.

Comparing the results of figure 22 with figure 3.8 [25] shows quite good agreement. However, exactly how well the models agree is hard to say since the results in [25] are only presented as coverage diagrams. Lindquist does however state that the path loss for the 5 GHz source at 200km is between $130 - 140$ dB which corresponds well with results obtained in this thesis, see figure 23b. The agreement should be good between the two since the devel-

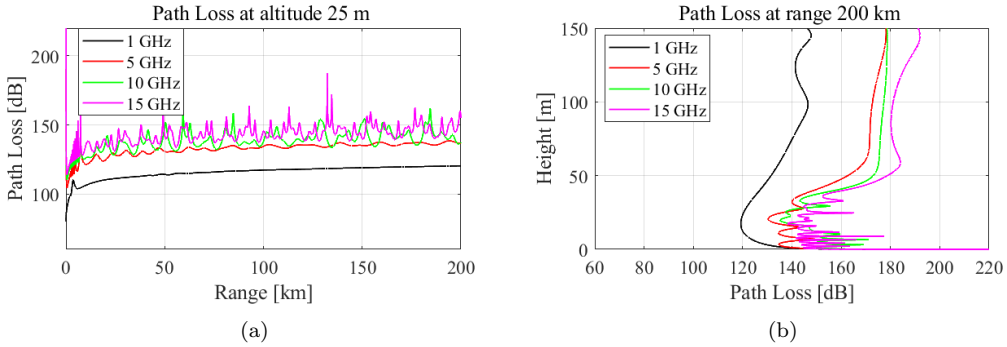


Figure 23: Path loss plots for figures 22a-22d. a) Path loss as function of range 25 m above the surface. b) Path loss as a function of height 200 km away from the source.

oped model for a PEC surface is based on the work of Lindquist. However, small changes to the antenna modelling was made which could alter the final results a bit. For an even better comparison a PETOOL v.2 simulation was performed with the same surface duct. The comparison between the developed model and PETOOL v.2 is shown in figure 24.

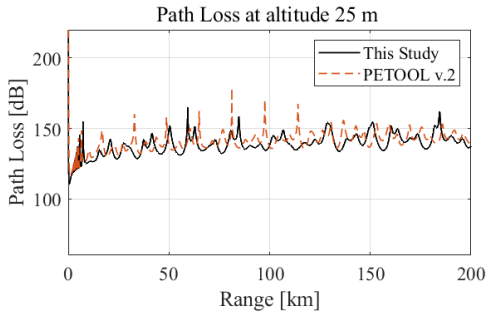


Figure 24: Comparison of the path loss vs range results obtained from developed model and PETOOL v.2.

The comparison in figure 24 shows that the path loss are of the same order and display similar behaviour.

4.2.2 Range Varying surface duct

This subsection concerns propagation in range varying atmospheric conditions. In figure 25 two coverage diagrams of the path loss and one plot comparing the path loss at a distance 200 km from the antenna are shown. For all simulations a horizontally polarized antenna placed 20 m above the surface was used. The ground was modelled as a PEC. In figure 25a the atmosphere goes from being standard at the range $x = 0$ to a 40 m heigh and 20 M-units strong surface duct at $x = 40$ km. It then remains constant through the rest of the domain. An almost opposite scenario is shown in figure 25b where the atmosphere starts as a surface duct and remain constant until $x = 160$km. The atmosphere then gradually transforms into standard atmosphere. Comparison between the figures shows a clear difference. When propagation starts in the standard atmosphere the field is initially bent upwards. Once the ducting layer gets strong enough the field that has not been bent far enough gets trapped into the ducting layer where it gets trapped. If the antenna is placed inside the ducting layer, the energy is confined inside the ducting layer. Once the duct strength decreases the duct can not confine the fields and therefore the fields starts to propagate as in standard atmosphere. The path loss as a function of height 200 km away from the source for both types of atmosphere is shown in figure 25c, together with results obtained from a PETOOL v.2 simulation where the atmosphere used can be found in table A4. The agreement between the results are very good inside the ducting layer, both models estimate the path loss inside the ducting layer to $140 - 160$ dB. Once outside the ducting layer PETOOL v.2 and results from this study do not match at all. This is probably due to the environmental propagator mentioned previously. What is interesting is that when comparing the results from figure 25c to figure 23b we see that when propagation starts in standard atmosphere the path losses are not that different. In fact, the minimum path losses are nearly the same.

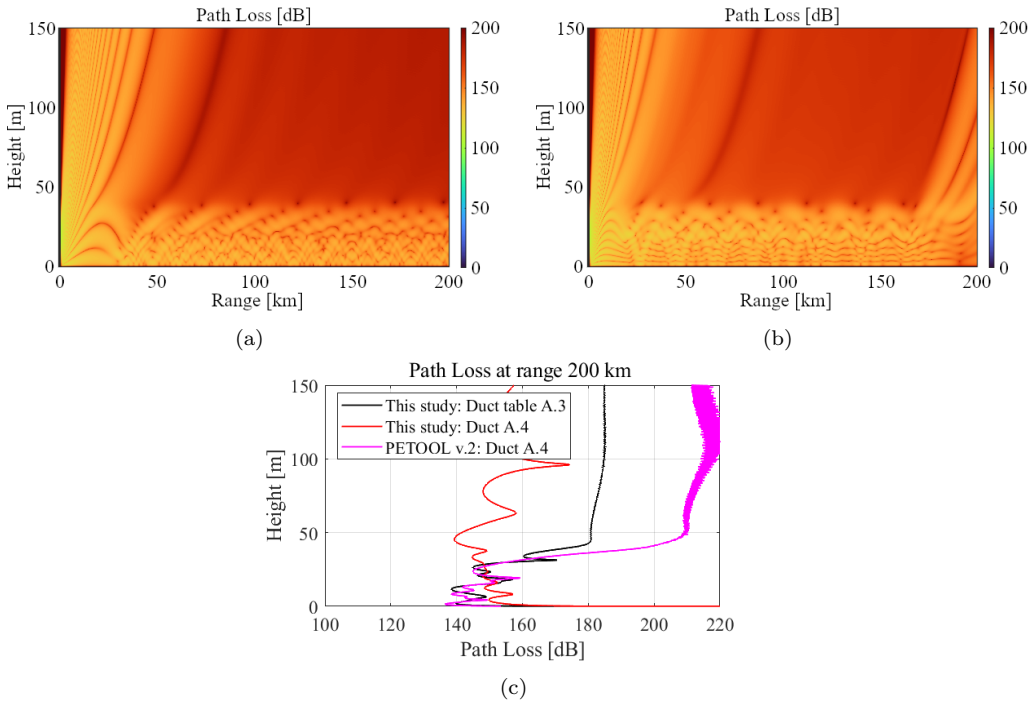


Figure 25: Propagation in a range varying atmosphere for a field with horizontal polarization. The atmosphere at some given points can be found in A4. a) Atmosphere going from standard atmosphere at to surface duct b) Atmosphere going from a surface duct to standard atmosphere

4.3 Terrain

In this part, wave propagation over terrain will be presented. Results from one simulation with the intension of reproducing figure 6 in [48] will be shown and compared to both GO+UTD (a program based on Geometrical Optics and Unified theory of diffraction) and PETOOL v.2.

The simulation uses a horizontally polarized antenna placed 120 m above the ground. Between the ranges $0-3$ km the terrain 0 m in height, between $3-4$ km the terrain height is linearly increases up to 100 m before it linearly decrease back to 0m at 5 km. Beyond 5 km the terrain height is constantly zero. The coverage diagram in figure 26a shows how the field is reflected of the pyramid and redirected upward. Beyond the pyramid, diffraction of the top causes interference between diffracted, reflected and incident waves

which gives rise to the interference pattern in the bottom right part in coverage diagram and the weakening of the field seen in the right upper half portion of the coverage diagram.

Figure 26b shows the propagation factor as function of range at an altitude of 50 m. The results are compared to PETOOL v.2 and G0+UTD and is a reproduction of figure 6 in [48]. First thing to notice is that since the ground is modelled as a PEC no energy penetrates the ground. Therefore the field is zero in the middle where the wedge is positioned. The agreement with PETOOL is very good except between $1.5 - 3.5$ km in where the location of the maximums and the depth of minimums differ. In figure 26c the propagation factor as a function of height 6 km from the antenna is shown. In the lower part of the figure the interference pattern caused by the diffraction can be seen.

The developed and PETOOL v.2 shows excellent agreement meanwhile comparison between this study and GO+UTD shows that the GO+UTD overestimates field strengths behind objects compared to the SSPE. This i clearly seen in the bottom right part of 26c and bottom part of 26b where the propagation factor is largere for GO+UTD. However, it is not surprising that the results from this thesis and GO+UTD differs since they are based on two different approaches. Nonetheless it is comforting to know that PETOOL v.2 and this study coincides when both are based on the SSPE.

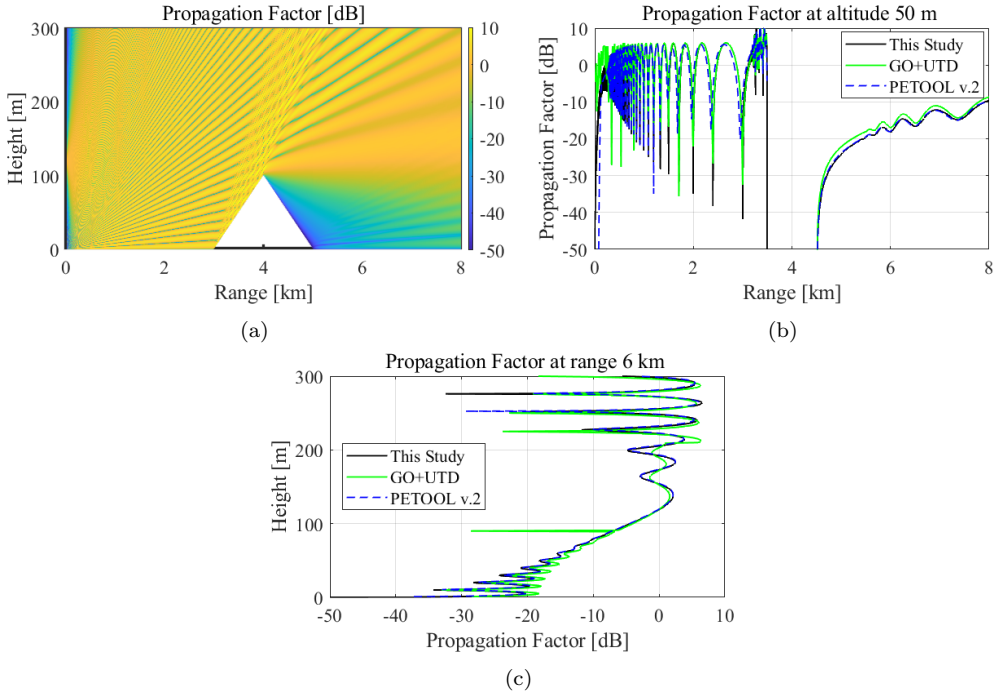


Figure 26: Propagation over a wedge terrain for a field with horizontal polarization. a) Coverage diagram of the propagation factor. The white corresponds to the terrain. b) Propagation factor as function of range 50 m above the surface. c) Propagation factor as function of height 6 km way from the source. The results obtained in this study is compared to G0+UTD and PETOOL v.2 in b) and c).

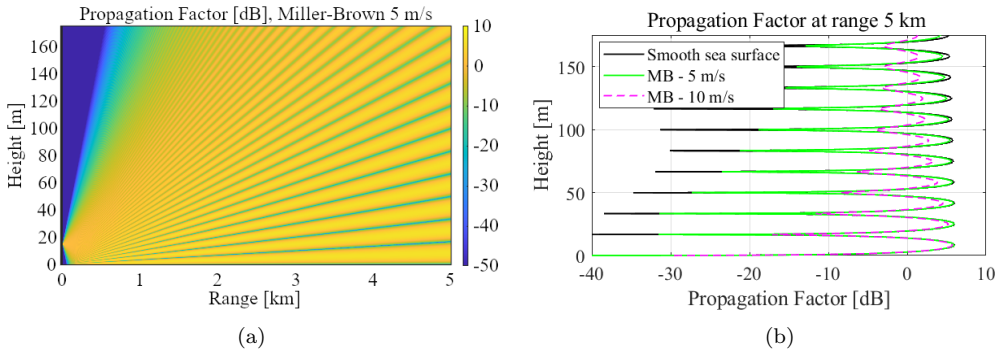
4.4 Sea surface

In this subsection wave propagation over different types of modelled sea surfaces are investigated. All the simulation uses impedance boundary conditions. The sea surface can either be considered to be flat and smooth, as a rough sea surface and using the Miller-Brown roughness reduction factor to determine the surface impedance, as irregular terrain, or as rough sea surface consisting of long-wave and short-wave roughness features. Multiple simulation are performed to test all of the above mentioned features. The First simulation concerns flat, smooth and rough sea surfaces and the results are compared to figure 3 in [49]. The second simulation concerns sea waves modelled as irregular terrain. This results will be discussed and not compared to any other source. The last set of simulations regards propagation in

surface ducts where the underlying terrain is modelled as different kinds of sea surfaces. The results from the last set of simulation will not be compared to a specific source, instead the different results are compared to different sources or non at all.

The first set of simulation concerns propagation over a rough sea surfaces. In figure 27a a coverage diagram of the propagation factor over a terrain modelled with the Miller-Brown method at wind speed 5 m/s is shown. In figure 27b the propagation for wind speed, 5 m/s, and 10 m/s is compared to the a smooth sea surface. For the rough sea surfaces the maximum and minimum, decreases and increases with height, respectively. This is true for the smooth sea surface as well, however, the decrease of maximum is so small it is barely visible in figure 27a. The decreased coherence with height for higher wind speed is to be expected since the Rayleigh roughness parameter given by equation (157) increases with the wind speed. The increased surface roughness causes specular scattering to be reduced and the coherence between direct and reflected paths is therefore reduced [49]. This results in gradual weakening of minimums and maximums with height. The agreement with figure 3 in [49] is very good. Notice that in [49] the authors have chosen elevation angle as y-axis instead of altitude, however, 2° of elevation corresponds to roughly 175m in altitude 5 km from the source. The results show the same general behaviour. The maximums and minimums are located at similar positions. However, in the results from this study the maximums are slightly lower in value and minimums are not as deep as in [49]. This is probably due to the fact that different sea spectrum models was used. This results in a difference between the root mean square height deviations of the surface h. which consequently affect the sea surface roughness through the Rayleigh roughness parameter.

The results from propagation simulation over sea surfaces modelled as irregular terrain can be seen in 28. Each sea surface is for a different wind speed. It is clear that the wind speed has an great impact on the propagation. In figure 28a the propagation over a sea surface for wind speed is 3m/s shown. The coverage diagram shows a clear interference pattern and it is almost as the sea surface does not affect the propagation. Considering that the sea surface barely is visible in the coverage diagram, this makes sense. For a wind speed of 7 m/s seen in figure 28b, the height of sea waves has increased. The interference pattern is still visible, however due to reflections



(a) (b)

Figure 27: Propagation over a rough sea surface with wind speed 5 m/s. a) Coverage diagram of the propagation factor obtained using the developed model. Propagation factor versus height for propagation over rough surfaces using the Miller-Brown (MB) method with wind speed 5 m/s, 10 m/s, and a smooth flat surface.

and diffraction off the surface waves the interference has diminished. The effect of the waves when the wind speed is 10 m/s is shown in 28c. The waves are even higher and thus the reflection and diffraction off them is increased. This leads to the interference pattern being further diminished. The propagation factor as a function of height 2 km away from the source is shown in figure 28d where the diminishing of the interference pattern is even more comprehensible. The lobe pattern for the 3m/s surface resembles that of the smooth sea surface in figure 27b. As the wind speed increases, the lobe pattern gets distorted with maximums, and minimums that decreases, and increases, respectively. Furthermore, as the wind speed increases the maximums and minimums becomes displaced. The same behaviour was noticed by [46]. The results obtained in this study shows similar behaviour as the results in [46]. However, it is hard to say if the results are similar since the notation and sea surface spectrum is different. Ehn denotes the sea surfaces by sea states, rather than wind speed which makes the comparison a bit harder.

Next, results from simulation regarding propagation in surface ducts over the different sea surface models will be presented. In figure 29a the sea surface is modelled as flat and smooth. The coverage diagram is very similar to that of figure 22c. Even the Path loss as function of range at 200 km, shown in figure 29f is almost identical to that of figure 23b. In figures 29b

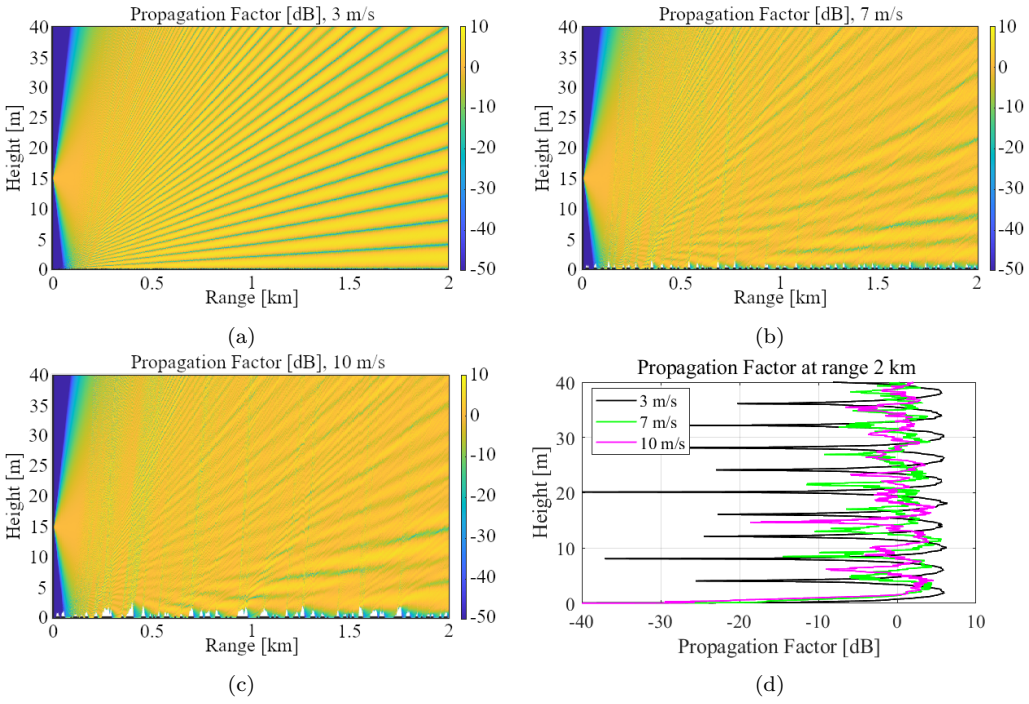


Figure 28: Propagation over sea surface generated through equation (172). a) Coverage diagram of propagation factor for wind speed 3 m/s. b) Coverage diagram of propagation factor for wind speed 7 m/s. c) Coverage diagram of propagation factor for wind speed 10 m/s. Propagation factor as function of height for the different sea surfaces.

and 29c the surface is modelled as a rough sea surface via the Miller-Brown method with wind speeds of 5 m/s and 10 m/s respectively. For the case of 5 m/s wind speed the path loss is reduced a bit. However, when looking the wind speed is increased to 10 m/s the path loss drastically increased with range. This is similar to the results shown in figure 27b and is once again due to the increased roughness of the surface. At a distance 200 km the difference between path loss for the two cases are $20 - 25$ dB. In figure 29d the sea surface has been modelled as irregular terrain. Inside the ducting layer the path loss is similar to the smooth surface as seen in figure 29f. The difference between the two is found outside the ducting layer where path loss curve becomes noicy. Waves that are reflected off the surface waves with large angles are not always going to remain inside the ducting layer, therefore some parts of the fields escape the ducting layer which then makes the path

loss noisy. However, it is possible that level of noice is due to a poorly performing method. When the DMFT was used for long range simulations the model failed when too small height increments were used. Therefore height increments were increased which could have possible made the overall performance worse. The last simulation used the full sea surface where large-scale rougness's is modelled as irregular terrain and the small-scale roughness's is modelled through the Miller-Brown method. The result shows an overall increment of path loss with range. From figure 29f we see that the full sea model has the highest path loss except for the rough surface at wind speed 10 m/s. Similar results were found by Benhmammouch et.al in [50] where the largest path loss was found for the full sea surface model.

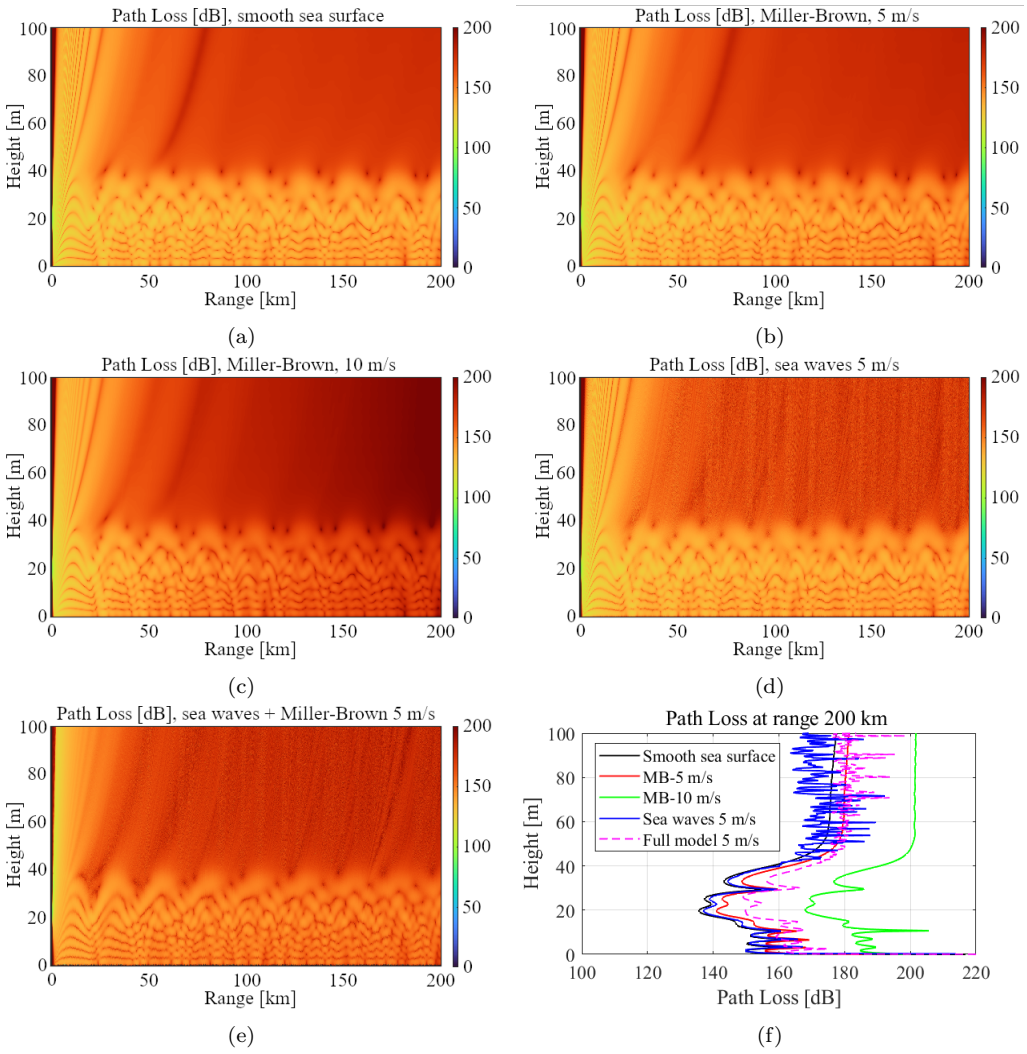


Figure 29: Path loss coverage diagrams for propagation over sea surfaces in a bilinear surface duct described in table A2. a) Smooth flat sea surface, b) Rough sea surface for wind speed 5m/s, c) Rough sea surface for wind speed 10m/s and d) Generated sea surface for wind speed 5m/s. e) Full sea wave model for wind speed 5m/s. f) Path loss versus height for the different sea surfaces.

4.5 The Model

The features of the developed model have been tested and compared to literature with good results. However, the model does have its problems. For

instance, the DMFT algorithm that was chosen to model non-PEC materials is very slow. The SSFT method is used due to its speed and with the DMFT that advantage is gone. Furthermore, the DMFT algorithm was not as numerically stable as the PEC algorithm and could not handle small height increments. The reason for this is unknown. The DMFT algorithm is not as intuitive as the PEC algorithm and it is possible that some errors causing numerical stability have slipped in. Another possibility for numerically instabilities could be bad alpha problem which are caused when $\Re(\alpha)$ is very small and $\Im(\alpha)$ is less than p_{max} [38]. This was for example found when using the DMFT together with the rough surface model. Than similar Big V problems as in [39] could be spotted.

Even though the method for handling terrain showed great agreement with PETOOL v.2 and GO+UTD the use of PLSM could have led to unrealistic results. The constant redirecting of the fields inside the duct could have led to unrealistic result in figures 29d and 29e. A better approach would have been to use the staircase method for duct simulations which does not require the fields to be redirected. Furthermore, the PLSM are best suited for terrain with inclination angles below 15°. When the wind speed is high the sea waves increases in size and their is possible that some inclinations was steeper than 15°.

The full sea surface model also has it drawbacks. Modelling the sea surface is in general a good approach since it effects that only can arise due to terrain to be accounted for. For example shadowing. However, their are a few different methods for modelling terrain. Nearly all of them are based on sea surface spectrum and it is hard to determine which one that best represent the reality. Furthermore, it is hard to determine how accurate the splitting of the roughness into large-scale and small-scale actually is.

The range independent environments showed promising results. However, the developed model could not handle both rough sea surfaces and varying refractivity. This was due to the calculation of grazing angles. The time delimitations on this thesis made it hard to develop a ray tracing method that could handle range varying refractive index. Therefore all the features of the model was never tested together.

Lastly, even though the duct strengths and heights that were used gives a good overlook of anomalous propagation they are maybe not the most realistic. Occasionally the duct strength and height can reach those values

but most often they are a little bit smaller.

5 Conclusion

This thesis has presented a model that can be used to estimate electromagnetic propagation over-seas and in a range varying atmosphere. The model is based on the parabolic equation and is discretized via the split-step Fourier transform method. For a PEC material a fast fourier transform approach was used and for non-PEC material the DMFT was used. The model was verified by comparing the obtained results to literature. The developed model showed good agreement with the results found in literature.

The range-dependent medium was modelled with a linear interpolation from tables of the M vs z profile. When propagating from a standard atmosphere into a surface the path loss was similar to that of plain duct propagation. The opposite scenario showed and increased path loss and energy leakage once the surface duct started to diminish.

Different sea surface was tested and it could be shown that for a flat and smooth surface, the propagation factor and the path loss was almost the same as for a PEC surface. When roughness effects were introduced through the Miller-Brown roughness reduction factor, the reflections off the surface increased which caused coherence to diminish. Modelling the sea surface as irregular terrain using the Elfouhaily showed that at moderate wind speeds the surface could almost be considered as flat. For higher wind speeds the reflection, refraction and diffractions off the surface increased and causing the interference patterns to diminish. Inside ducts the Miller-Brown method showed the path loss is significantly increased when high wind speeds were tested. The full sea model was able to take into account both small and large-scale roughness's. The full sea model had the highest path loss in atmospheric duct for wind speed of 5 m/s. However, the full sea model should be tested by using the staircase method of modelling terrain to further verify the results.

The results from simulations showed that path losses for waves propagating over a sea surface was in the order of $140 - 160$ dB for a wind speed of 5 m/s. Thus, it is plausible to achieve long range radio communication links through ducting channels by using signals with frequency of $1 - 15$GHz.

6 Future Work

In the list below a few proposals of possible future work that can be integrated with the current model.

- It would be interesting to look at hybrid approaches for terrain modelling. The Piecewise Linear Shift Map used for modelling the terrain in this work is only valid as long as the slopes are below 10°. However, programs like TEMPER uses hybrid approaches in which a mix between the staircase terrain masking approach and PLSM is used depending on the slope of the terrain.

- When investigating propagation over rough sea surfaces the grazing angles must be estimated. This is either done through a spectral estimator or geometrical optics/ray tracing methods. In this work a ray tracing program was used. However, RT is not the best choice for determining grazing angles when irregular terrain is present and therefore the use of a spectral estimator for determining the grazing angles could be investigated.

- It would be interesting to do a comprehensive study of evaporation duct models. Including them into the model would lead to further increasing the accuracy of the model.

References

[1] IEEE Antennas and Propagation Society. IEEE Standard Definitions of Terms for Radio Wave Propagation. Technical report, 2018.

[2] Ergin Dinc and Ozgur B. Akan. Beyond-line-of-sight communications with ducting layer. *IEEE Communications Magazine*, 52(10):37–43, 2014.

[3] R. K. Sharma, Purnima K. and Sharma, Dinesh and Singh. *Development of Field Propagation Model for Urban Area.* Anchor Academic Publishing, Hamburg, 2017.

[4] Jue Wang, Haifeng Zhou, Ye Li, Qiang Sun, Yongpeng Wu, Shi Jin, Tony Q.S. Quek, and Chen Xu. Wireless channel models for maritime communications. *IEEE Access*, 6:68070–68087, 2018.

[5] Arafat Habib and Sangman Moh. Wireless channel models for over-the-sea communication: A comparative study. *Applied Sciences (Switzerland)*, 9(3):1–32, 2019.

[6] Ergin Dinc and Ozgur B. Akan. Channel model for the surface ducts: Large-scale path-loss, delay spread, and AOA. *IEEE Transactions on Antennas and Propagation*, 63(6):2728–2738, 2015.

[7] Ergin Dinc and Ozgur B. Akan. Correction to "Channel Model for the Surface Ducts: Large-Scale Path-Loss, Delay Spread, and AOA" [Jun 15 2728-2738]. *IEEE Transactions on Antennas and Propagation*, 64(8):3735–3736, 2016.

[8] ITU-R. Method for point-to-area predictions for terrestrial services in the frequency range 30 MHz to 3 000 MHz. *Recommendation ITU-R P.1546-5*, 4, 2013.

[9] Technical Specification, Group Radio, Access Network, and Physical Layer Aspects. 3Gpp TR 38.802. 0(Release 14), 2017.

[10] Han Wang, Wencai Du, and Xing Chen. Evaluation of radio over sea propagation based ITU-R recommendation P.1546-5. *Journal of Communications*, 10(4):231–237, 2015.

[11] Herbert V. Hitney, Juergen H. Richter, Richard A. Pappert, Kenneth D. Anderson, and Gerald B. Baumgartner. Tropospheric Radio Propagation Assessment. *Proceedings of the IEEE*, 73(2):265–283, 1985.

[12] Graham S Woods, Adam Ruxton, Cameron Huddlestone-holmes, and Gilles Gigan. High-Capacity, Long-Range, Over Ocean Microwave Link Using the Evaporation Duct. *IEEE Journal of Oceanic Engineering*, 34(3):323–330, 2009.

[13] Asif Iqbal and Varun Jeoti. Feasibility study of radio links using evaporation duct over sea off Malaysian shores. *2010 International Conference on Intelligent and Advanced Systems, ICIAS 2010*, 10(2):2–6, 2010.

[14] Mamta Agiwal, Abhishek Roy, and Navrati Saxena. Next generation 5G wireless networks: A comprehensive survey. *IEEE Communications Surveys and Tutorials*, 18(3):1617–1655, 2016.

[15] Amalia E Barrios. Parabolic Equation Modeling in Horizontally Inhomogeneous Environments. *IEEE Transactions on Antennas and Propagation*, 40(7):791–797, 1992.

[16] Mireille Levy. *Parabolic equation methods for electromagnetic wave propagation*. IET, London, 2000.

[17] Irina Sirkova. Brief review on PE method application to propagation channel modeling in sea environment. *Central European Journal of Engineering*, 2(1):19–38, 2012.

[18] T. Elfouhaily, B. Chapron, K. Katsaros, and D. Vandemark. A unified directional spectrum for long and short wind-driven waves. *Journal of Geophysical Research C: Oceans*, 102(C7):15781–15796, 1997.

[19] Othmane Benhmammouch, Ali Khenchaf, and Natacha Caouren. Modelling Roughness effects on propagation of electromagnetic waves in a maritime environment: a hybrid approach. *IET Radar, Sonar and Navigation*, 5(9):1018–1025, 2011.

[20] Wenwan Ding, Kun Wang, and Yunliang Long. Study of the Electromagnetic Waves Propagation over the Improved Fractal Sea Surface Based on Parabolic Equation Method. *International Journal of Antennas and Propagation*, 2016, 2016.

[21] Amalia E Barrios. Considerations in the Development of the Advanced Propagation Model (APM) For U.S Navy Applications. *Assisted Circulation 4*, pages 77–82, 1995.

[22] G. Daniel Dockery, Ra'id S. Awadallah, David E. Freund, Jonathan Z. Gehman, and Michael H. Newkirk. An overview of recent advances for the TEMPER radar propagation model. *IEEE National Radar Conference - Proceedings*, pages 896–905, 2007.

[23] Ozlem Ozgun, Gökhan Apaydin, Mustafa Kuzuoglu, and Levent Sevgi. PETOOL: MATLAB-based one-way and two-way split-step parabolic equation tool for radiowave propagation over variable terrain. *Computer Physics Communications*, 182(12):2638–2654, 2011.

[24] Ozlem Ozgun, Volkan Sahin, Muhsin Eren Erguden, Gokhan Apaydin, Asim Egemen Yilmaz, Mustafa Kuzuoglu, and Levent Sevgi. PETOOL v2.0: Parabolic Equation Toolbox with evaporation duct models and real environment data. *Computer Physics Communications*, 256:107454, 2020.

[25] Tim Lindquist. *Wave Propagation Models in the Troposphere for Long-Range UHF / SHF Radio Connections*. Master thesis, Karlstad University, 2020.

[26] David J Griffiths. *Introduction to electrodynamics*. Cambridge University Press, Cambridge, 4 edition, 2017.

[27] John David Jackson. *Classical Electrodynamics*. Wiley, 3 edition, 1999.

[28] Donald E. Kerr. *Propagation of Short Radio Waves*. McGraw-Hill Book Company, New York, m.i.t radiation laboratory series, volume 13 edition, 1951.

[29] Pei Zhang, Lu Bai, Zhensen Wu, and Lixin Guo. Applying the Parabolic Equation to Tropospheric Groundwave Propagation: A review of recent achievements and significant milestones. *IEEE Antennas and Propagation Magazine*, 58(3):31–44, 2016.

[30] R.H Hardin and F.D Tappert. Applications of the split-step Fourier method to the numerical solution of nonlinear and variable coefficient wave equations. *SIAM Review*, 15(2):423, 1973.

[31] M. D. Feit and J. A. Fleck. Light propagation in graded-index optical fibers. *Applied Optics*, 17(24), 1978.

[32] D. J. Thomson and N. R. Chapman. A wide-angle split-step algorithm for the parabolic equation. *Journal of the Acoustical Society of America*, 74(6):1848–1854, 1983.

[33] Mark J Ablowitz and Athanassios S Fokas. *Complex Variables : Introduction and Applications*. Cambridge University Press, 2 edition, 2003.

[34] Frank J Ryan. Technial Report 1453: Analysis of Electromagnetic Propagation Over Variable Terrain Using The Parabolic Wave Equation. Technical report, 1991.

[35] Harvey W. Ko, James W. Sari, and Joseph P. Skura. Anomalous Microwave Propagation Through Atmospheric Ducts. *Johns Hopkins APL Technical Digest (Applied Physics Laboratory)*, 4(1):12–26, 1983.

[36] International Telecommunication Union. Recommendation ITU-R P.527-5: Electrical characteristics of the surface of the Earth:. Technical report, 2019.

[37] James R Kuttler and G. Daniel Dockery. Theoretical description of the parabolic approximation/Fourier split-step method of representing electromagnetic propagation in the troposphere. *Radio Science*, 26(2), 1991.

[38] G. Daniel Dockery and James R. Kuttler. An Improved Impedande-Boundary Algorithm for Fourier Split-Step Solution of the Parabolic Wave Equation. *IEEE Transactions on Antennas and Propagation*, 44(12):1592–1599, 1996.

[39] James R. Kuttler and Ramakrishna Janaswamy. Improved Fourier transform methods for solving the parabolic wave equation. *Radio Science*, 37(2):5–1–5–11, 2002.

[40] Radiocommunication Bureau. *Handbook on Radiometeorology*. 2014.

[41] B R Bean and E J Dutton. *Radio Meteorology*. National Bureau of Standards, Washington, national bureau of standards edition, 1966.

[42] International Telecommunication Union. Recommendation ITU-R P.453-14: The radio refractive index: its formula and refractivity data. Technical report, 2019.

[43] Denis J. Donohue and J. R. Kuttler. Propagation modeling over terrain using the parabolic wave equation. *IEEE Transactions on Antennas and Propagation*, 48(2):260277, 2000.

[44] A.R Miller, R.M Brown, and E Vegh. New derivation for the rough-surface reflection coefficient and for the distribution of sea-wave elevations. *IEE Proceedings H - Microwaves, Optics and Antennas*, 131(2):114–116, 1984.

[45] Ralph O. Schmidt. Multiple Emitter Location and Signal Parameter Estimation. *IEEE Transactions on Antennas and Propagation*, 34(3):276–280, 1986.

[46] Jonas Ehn. *Propagaton of Radio Waves in a Realistic Environment using a Parabolic Equation Approach.* Master thesis, Linköping University, 2019.

[47] G. Daniel Dockery. Modeling Electromagnetic Wave Propagation in the Troposphere Using the Parabolic Equation. *IEEE Transactions on Antennas and Propagation*, 36(10):1464–1470, 1988.

[48] Ozlem Ozgun. New Software Tool (GO+UTD) for Visualization of Wave Propagation [Testing Ourselves]. *IEEE Antennas and Propagation Magazine*, 58(3):91–103, 2016.

[49] D. E. Freund, N. E. Woods, H. C. Ku, and R. S. Awadallah. The effects of shadowing on modelling forward radar propagation over a rough sea surface. *Waves in Random and Complex Media*, 18(3):387–408, 2008.

[50] Othmane Benhmammouch, Natacha Caouren, and Ali Khenchaf. Influence of Sea Surface Roughness on Electromagnetic Waves Propagation in Presence of Evaporation Duct. In *Radar Conference - Surveillance for a Safer World, 2009. RADAR. International*, 2009.

Appendices

A

Table A1: Simulation parameters that have been used to obtain the results presented in Section 4. The columns are: Maximum range, increment step in x, maximum height,increment step in z, frequency, antenna height, half power beam width, polarisation and type of atmosphere, ground, and terrain

Figure:	X_{max} [km]	Δx [m]	Z_{max} [m]	Δz [m]	f [GHz]	a_h [m]	β [°]	**Atm.:**	**Pol.:**	**Gro.:**	**Terr.:**
Figure 14	10	10	100	1/32	3	50	8	Vac.	Hor.	PEC	None
Figure 15	10	10	100	1/32	3	50	8	Vac.	Ver.	PEC	None
Figure 16-17	100	50	350	1/32	3	50	3	Sta.	Hor.	PEC	None
Figure 18	100	50	350	1/32	3	31	3	Sta.	Hor.	PEC	None
Figure 19	100	50	1200	1	1	25	3	Sta.	Hor.	PEC	None
Figure 20-21	100	50	350	1/32	3	50	3	Sta.	Ver.	PEC	None
Figure 22-24	200	20	150	1/32	1-15	20	1	Table A2	Hor.	PEC	None
Figure 25	200	20	150	1/32	10	20	1	Table A4 & A1	Hor.	PEC	None
Figure 26	8	1	300	1/32	0.3	120	45	Vac.	Hor.	PEC	Table A5
Figure 27	5	10	175	1/12	3	15	8	Vac.	Hor.	Sea Water	MB
Figure 28	2	0.5	40	1/32	5	15	8	Vac.	Hor.	Sea Water	Sea Waves
Figure 29	2	10	100	1/12	10	20	1	Table A2	Hor.	Sea Water	Sea Waves & MB

Atm-Atmosphere, Vac-Vacuum, Sta-Standard, Pol-Polarization, Hor-Horizontal, Ver-Vertical, Gro-Ground material, Terr-Terrain, MB-Miller-Brown.

Table A2: The M values that was used to describe the duct in figures 22 and 29

M-Value [M-units]	Altitude [m]
330	0
310	40
322.87	150

Table A3: The M values that was used to describe the duct in figure 25b

Range [km]	M-Value [M-units]	Altitude [m]
0	330	0
	310	40
	322.87	150
160	330	0
	310	40
	322.87	150
200	330	0
	347.55	150

Table A4: The M values that was used to describe the duct in figure 25a

Range [km]	M-Value [M-units]	Altitude [m]
0	330	0
	347.55	150
40	330	0
	310	40
	322.87	150
200	330	0
	310	40
	322.87	150

Table A5: Range vs Height values that used to obtain the terrain profile in 26

Range [km]	Altitude [m]
0	0
3	0
4	100
5	0
8	0